

On the Decidability of $\mathcal{ALCI}_{RCC5}$: A Survey of the Problem, Its History, and a Proposed Solution

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Abstract

The description logic $\mathcal{ALCI}_{RCC5}$ extends $\mathcal{ALCI}$ with the five RCC5 base relations as roles, under composition-table semantics. Its decidability has been open since Wessel (2002/2003). We survey the history of the problem—from Cohn’s original proposal (1993) through Wessel’s systematic investigation and Lutz & Wolter’s related undecidability results (2006)—and explain why all known undecidability reductions fail for $\mathcal{ALCI}_{RCC5}$. We then present a proposed decision procedure: the *cover-tree tableau*, which decomposes models into PPI-oriented cover trees where only $\{DR, PO\}$ remain as open cross-edges. This decomposition, combined with the patchwork property of RCC5, yields finite blocking via rank- d signatures. The approach is supported by extensive computational evidence (911 test concepts, zero mismatches; 775/775 SAT concepts with cover-tree models) but has not been formally verified by human experts. We discuss why naïve tableau approaches fail for $\mathcal{ALCI}_{RCC5}$ and how the cover-tree tableau overcomes these obstacles.

Keywords

description logic, spatial reasoning, RCC5, decidability, tableau calculus, patchwork property

1. Introduction

The decidability of concept satisfiability in the spatially extended description logic $\mathcal{ALCI}_{RCC5}$ has been an open problem for over twenty years. This paper provides a self-contained overview of the problem, its intellectual history, the obstacles that have resisted solution, and a proposed decision procedure that—if correct—would settle the question positively.

The paper is organized as follows. Section 2 traces the history of $\mathcal{ALCI}_{RCC5}$ from Cohn’s original 1993 proposal through Wessel’s systematic investigation and Lutz & Wolter’s topological undecidability results. Section 3 explains why standard undecidability reductions fail, including a concrete domino-encoding attempt that we verify is blocked. Section 4 discusses why naïve tableau approaches fail for complete-graph logics. Section 5 presents the split-forest model decomposition. Section 6 describes the cover-tree tableau and explains how it achieves blocking. Section 8 concludes with an assessment of what has been achieved and what remains to be done.

Disclaimer and AI-tool disclosure. This paper was prepared by Michael Wessel with significant assistance from generative-AI language tools: Claude (Anthropic, Opus 4.7 and Opus 4.8) [48], GPT-5.4 Pro (OpenAI) [49], and GPT-5.5 (OpenAI) [16]. The AI tools were used to formalise the mathematical proofs, implement the cover-tree tableau decision procedure, conduct the computational verification campaigns, perform the cold reviews that found successive defects, and draft the manuscript text including this overview. Wessel reviewed and edited the content as needed, directed the research, identified and corrected errors, and takes full responsibility for the publication’s content—including any errors, mistakes, or misleading statements the AI tools may have introduced. See the Declaration on Generative AI at the end of the paper for further details. The results and proofs have **not been peer-reviewed or verified by human domain experts**; they are published as a discussion piece, and the one completeness gap a cold review found (D-1) is repaired at the manuscript level in round-9 (§7.9) but not independently certified. The claims should not be taken as established results unless independently verified. Scrutiny, corrections, and feedback are invited.

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2. History of $\mathcal{ALCI}_{RCC5}$

2.1. Cohn’s 1993 proposal

The idea of combining description logics with qualitative spatial reasoning via modal accessibility relations was first proposed by Cohn [1], who considered treating the RCC8 relations (or subsets thereof) as modalities in a multi-modal logic. The framework allowed spatial constraints to be expressed as modal formulas—e.g., $\langle PP \rangle C$ for “some proper part satisfies C ”—but Cohn left all decidability and complexity questions open.

2.2. Wessel’s $\mathcal{ALCI}_{RCC}$ family (2002/2003)

Wessel [7, 8, 5] gave the first rigorous treatment of the combination, introducing the $\mathcal{ALCI}_{RCC}$ family: description logics $\mathcal{ALCI}_{RCCk}$ ($k = 1, 2, 3, 5, 8$) extending $\mathcal{ALCI}$ with role boxes derived from the RCC composition tables. In an $\mathcal{ALCI}_{RCCk}$ interpretation, every pair of domain elements is assigned exactly one RCC k base relation, and the composition table is enforced globally (*strong EQ semantics*: EQ is identity). Models are complete graphs with RCC k edge-colorings.

Wessel established the following:

- **Decidable cases:** $\mathcal{ALCI}_{RCC1}$, $\mathcal{ALCI}_{RCC2}$, and $\mathcal{ALCI}_{RCC3}$ are decidable, because their composition tables are deterministic or near-deterministic, admitting reduction to standard DL techniques.
- **Lower bounds:** $\mathcal{ALCI}_{RCC5}$ is PSPACE-hard; $\mathcal{ALCI}_{RCC8}$ is EXPTIME-hard.
- **Open:** The decidability of $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$ was left as the central open problem.

In a series of companion reports, Wessel also mapped out the decidability landscape for description logics with general composition-based role inclusion axioms:

- $\mathcal{ALC}_{\mathcal{RA}}$ (arbitrary role axioms) is **undecidable**, via reduction from the Post Correspondence Problem [6].
- $\mathcal{ALC}_{\mathcal{RA}}^{\ominus}$ (non-disjoint-roles variant) is also **undecidable**, via reduction from non-empty CFG intersection [3].
- $\mathcal{ALC}_{\mathcal{RASG}}$ (admissible role boxes: deterministic, functional, complete, and associative) is **decidable** (EXPTIME-complete, via reduction to $\mathcal{ALC}$ with general TBoxes) [4]; adding unqualified number restrictions ($\mathcal{ALCN}_{\mathcal{RASG}}$) makes it undecidable again.

$\mathcal{ALCI}_{RCC5}$ sits precisely in the gap: its role box is non-deterministic (unlike $\mathcal{ALC}_{\mathcal{RASG}}$) but has the patchwork property (unlike $\mathcal{ALC}_{\mathcal{RA}}/\mathcal{ALC}_{\mathcal{RA}}^{\ominus}$).

Remark 2.1 (Key difficulties). $\mathcal{ALCI}_{RCC5}$ has none of the properties that standard DL decidability proofs rely on: (i) no tree model property (models are complete graphs), (ii) no finite model property (some satisfiable concepts require infinite models), (iii) a non-deterministic role box (multi-valued composition entries).

For reference, the full RCC5 composition table is given in Table 1. Reading: row = $S(b, c)$, column = $R(a, b)$, entry = possible relations for (a, c) .

Key composition facts exploited in the cover-tree theory (Sections 5–6): $\text{comp}(PP, DR) = \{DR\}$ and $\text{comp}(DR, PPI) = \{DR\}$ (DR rigidity); $\text{comp}(PP, PP) = \{PP\}$ and $\text{comp}(PPI, PPI) = \{PPI\}$ (transitivity); and $\text{comp}(DR, PO) = \{DR, PO, PP\}$ (PP can be *forced* by composition and universal filtering; see Example 6.2).

Table 1

The RCC5 composition table (* = all five relations).

comp	DR(a, b)	PO(a, b)	EQ(a, b)	PPI(a, b)	PP(a, b)
DR(b, c)	*	DR, PO, PPI	DR	DR, PO, PPI	DR
PO(b, c)	DR, PO, PP	*	PO	PO, PPI	DR, PO, PP
EQ(b, c)	DR	PO	EQ	PPI	PP
PP(b, c)	DR, PO, PP	PO, PP	PP	PO, EQ, PP, PPI	PP
PPI(b, c)	DR	DR, PO, PPI	PPI	PPI	*

2.3. Wessel’s grid constructions and the coincidence obstruction

In his investigation of $\mathcal{ALCI}_{RCC8}$ [8], Wessel made two important discoveries:

1. Even-odd chains (Section 4.4.2). The TPP/NTPP distinction in RCC8 allows the construction of *functional successor chains*: each node alternates between *even* and *odd*, and a node can distinguish its direct TPPI-successor from all indirect ones by parity. Wessel used this to encode a binary n -bit counter, proving $\mathcal{ALCI}_{RCC8}$ is EXPTIME-hard.

2. The grid coincidence obstruction (Section 4.4.3, Figure 11). Wessel constructed explicit RCC8 networks isomorphic to $n \times n$ grids, using TPP for horizontal and vertical successors. However, he identified a fundamental obstacle to encoding domino tilings: the *coincidence condition* $R_X \circ R_Y = R_Y \circ R_X$ (east-of-north equals north-of-east) cannot be enforced by concept terms. The composition table allows $\{PO, EQ, TPP, NTPP, TPPI, NTPPI\}$ at the coincidence point—too many options. Wessel concluded:

“Even though these infinite models do exist, we have found no way to *enforce* the depicted model(s) by means of a concept term.”

Wessel further observed that adding a hybrid logic binding operator $\downarrow$ (nominals) would immediately yield undecidability, confirming that the coincidence condition is the *precise boundary*.

Wessel thus independently developed, three to four years before Lutz–Wolter, the RCC8 description-logic analogues of these ingredients—the even-odd chain, an explicit two-dimensional TPP/NTPP grid, and the first identification of the coincidence gap as the decisive obstacle. Lutz–Wolter later obtained undecidability by a *different* route—a linearized tiling enumeration (their §5, following Marx–Reynolds) that sidesteps the two-dimensional coincidence rather than enforcing it—so this is shared lineage and prior identification, not a dependence of their proof on Wessel’s constructions.

2.4. Lutz & Wolter (2006)

Lutz and Wolter [9] studied *modal logics of topological relations*, denoted L_{RCC8} , L_{RCC5} , etc. These have the same *syntax* as $\mathcal{ALCI}_{RCC}$ but are interpreted over *topological models*: the domain elements are regular closed subsets of a topological space (typically $\mathbb{R}^n$). They proved:

- L_{RCC8} is **undecidable** (their §5: a linearized NTPP-chain enumeration of tile positions, following Marx–Reynolds, whose faithfulness relies on the geometry of $\mathbb{R}^n$; no two-dimensional grid coincidence is enforced).
- $L_{RCC5}(RS^\exists)$ is **undecidable** (reduction from $S5^\exists$ via supremum regions).
- $L_{RCC5}(RS)$ is **open**. Lutz and Wolter explicitly wrote (p. 31): “Perhaps the most interesting candidate is $L_{RCC5}(RS)$ [...] to which the reduction exhibited in Section 8 does not apply.”

The logic $\mathcal{ALCI}_{RCC5}$ is exactly $L_{RCC5}(RS)$: arbitrary complete-graph models with composition-table semantics. The problem has been recognized as open by leading researchers since 2006.

Table 2

Why standard undecidability reductions fail for $\mathcal{ALCI}_{RCC5}$.

Source Problem	Technique	Missing in $\mathcal{ALCI}_{RCC5}$
$\mathbb{N} \times \mathbb{N}$ domino	Graded modalities + transitivity	Number restrictions
$\mathcal{ALC}_{\mathcal{RA}}$ [6]	Post Correspondence Problem	Arbitrary role axioms
$\mathcal{ALC}_{\mathcal{RA}}^{\ominus}$ [3]	Non-empty CFG intersection	Non-disjoint roles
$\mathcal{ALCI}_{RCC8}$ (Wessel)	Grid via TPP/NTPP	TPP/NTPP distinction
L_{RCC8} (Lutz–Wolter)	Linearized topological enumeration	Topological rigidity
$L_{RCC5}(RS^{\exists})$	$S5^3$ via supremum regions	Supremum closure

3. Why Undecidability Reductions Fail

Undecidability proofs for description logics use a variety of techniques: grid encoding ($\mathbb{N} \times \mathbb{N}$ domino tiling, requiring functional roles or number restrictions), reduction from the Post Correspondence Problem (PCP), reduction from context-free grammar (CFG) intersection, or simulation of $S5^3$ via geometric constructions. $\mathcal{ALCI}_{RCC5}$ lacks the features that each of these techniques exploits: it has no functional roles, no number restrictions, no arbitrary role box, and no geometric coincidence conditions.

3.1. Blocked reduction routes

Table 2 summarizes the blocked routes. The **patchwork property** of RCC5 (Renz & Nebel [2])—local (triple-wise) consistency implies global consistency for atomic networks—actively resists grid encoding, since tiling reductions require rigid global constraints that go beyond local consistency.

3.2. A concrete domino attempt in $\mathcal{ALCI}_{RCC5}$

Can we encode a domino-like grid using PP/PPI chains in RCC5? Consider the concept:

$$\begin{aligned}
 X = & \exists \text{PPI}.\exists \text{PPI}.C \sqcap \exists \text{PPI}.\exists \text{PPI}.D \\
 & \sqcap \forall \text{PPI}.\forall \text{PPI}.(\forall \text{PO}.(\neg C \sqcap \neg D) \sqcap \forall \text{DR}.(\neg C \sqcap \neg D)) \\
 & \qquad \sqcap \forall \text{PPI}.(\neg C \sqcap \neg D) \sqcap \forall \text{PP}.(\neg C \sqcap \neg D)) \\
 & \sqcap \forall \text{PPI}.\exists \text{PPI}.X
 \end{aligned}$$

The intent: each X -node has two PPI-grandchildren colored C and D (the “square”), and the universals isolate the colored nodes from all non-EQ neighbors. The recursive $\forall \text{PPI}.\exists \text{PPI}.X$ forces infinite repetition.

One level is satisfiable. Without the recursive conjunct, X has a 3-element model: $z \text{ PP } y \text{ PP } x$, with $z \in C \sqcap D$. The two grandchildren are forced to be *the same element* (EQ collapse): if $z_1 \neq z_2$, they would be related by some $R \in \{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$, and the universals would require $z_2 \notin D$ (firing z_1 ’s $\forall R.\neg C \sqcap \neg D$). So $z_1 = z_2 \in C \sqcap D$, and the universals are vacuously satisfied (the only neighbors of z are PP-predecessors, not PO/DR/PPI-neighbors).

Two levels is unsatisfiable. Adding $\forall \text{PPI}.X$: the PPI-child y must itself satisfy X (without recursion), producing a new $C \sqcap D$ -labeled element w as a PPI-child of z . But z is a PPI-grandchild of x , and x ’s universals fire: z ’s PPI-neighbor w must satisfy $\neg C \sqcap \neg D$. Yet $w \in C \sqcap D$. **Contradiction.**

This was verified computationally: a simplified version $X_1 = \exists \text{PPI}.\exists \text{PPI}.C \sqcap \forall \text{PPI}^3.\neg C$ is SAT; $X_2 = X_1 \sqcap \forall \text{PPI}.X_1$ is UNSAT. The domino encoding collapses at depth 2 because the isolation constraints from the grandparent level kill the colored elements created by the recursive unfolding.

Remark 3.1. The failure is not accidental. In $\mathcal{ALCI}_{RCC5}$, the only way to create “functional” successors is via PP/PPI chains, but $\text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$ makes these chains *transitive*—grandparent universals automatically propagate to grandchildren, preventing the positional diversity needed for grid encoding. In $\mathcal{ALCI}_{RCC8}$, the TPP/NTPP distinction breaks this transitivity (a direct TPPI-child is distinguishable from indirect descendants), which is why Wessel’s even-odd chain works there but has no RCC5 analogue.

3.3. A concrete domino attempt in $\mathcal{ALCI}_{RCC8}$

The remark above suggests the TPPI/NTPPI distinction in $\mathcal{ALCI}_{RCC8}$ might rescue the construction: a direct successor (TPPI) and an indirect one (NTPPI) are genuinely distinguishable, so perhaps the level-2 collapse can be avoided. We attempt the analogue and show that while it escapes the RCC5 failure mode, it fails for a different reason—Wessel’s *coincidence obstruction* [8].

The schema. Let Corner be the concept that isolates a node from all non-EQ neighbors:

$$\text{Corner} = \bigwedge_{R \in \{\text{DC}, \text{EC}, \text{PO}, \text{TPP}, \text{NTPP}, \text{TPPI}, \text{NTPPI}\}} \forall R. (\neg C \sqcap \neg D).$$

The isolation is placed *locally on the colored grandchildren* rather than as an ancestral $\forall \text{TPPI}. \forall \text{TPPI}$ universal, so it cannot propagate transitively through the recursion:

$$\begin{aligned} X = & \exists \text{TPPI}. \exists \text{TPPI}. (C \sqcap \text{Corner}) \sqcap \exists \text{TPPI}. \exists \text{TPPI}. (D \sqcap \text{Corner}) \\ & \sqcap \forall \text{TPPI}. (\neg C \sqcap \neg D) \sqcap \forall \text{TPPI}. X. \end{aligned}$$

The third conjunct $\forall \text{TPPI}. (\neg C \sqcap \neg D)$ forces the corner relation $x \rightarrow z$ to be NTPPI (not TPPI), using $\text{comp}(\text{TPPI}, \text{TPPI}) = \{\text{TPPI}, \text{NTPPI}\}$.

Level 1 is satisfiable, with a real direct/indirect distinction. Without recursion, x has TPPI-children y_1, y_2 (the two “direct” positions) and a single corner z reached by TPPI-TPPI chains from both y_1 and y_2 . The EQ-collapse at level 2 works exactly as in RCC5: $z_1 \in C \sqcap \text{Corner}$ and $z_2 \in D \sqcap \text{Corner}$ are mutual non-EQ neighbors only if Corner is violated, so under strong-EQ semantics $z_1 = z_2 = z$. Crucially, $z \neq y_1, y_2$ in general: z is NTPPI of x while y_1, y_2 are TPPI—the direct/indirect distinction survives, which is a genuine RCC8 refinement over the RCC5 attempt of §3.2.

Level 2 collapses via the coincidence obstruction. Adding $\forall \text{TPPI}. X$: the TPPI-child y of x satisfies X , producing its own corner $w \in C \sqcap D \sqcap \text{Corner}$ at NTPPI-distance from y , and hence at NTPPI-distance from x via $\text{comp}(\text{TPPI}, \text{NTPPI}) = \{\text{NTPPI}\}$. Now x ’s own corner z and y ’s corner w are distinct grid positions in the intended reading—but as elements of the model they are related by some RCC8 base relation. Since z carries Corner, every non-EQ neighbor of z lies in $\neg C \sqcap \neg D$. But $w \in C \sqcap D$. Therefore $z = w$ under strong-EQ. **All corners across the grid coincide at a single point.**

Why the TPPI/NTPPI trick fails. In the RCC5 attempt the collapse is a *transitive-propagation* failure: the grandparent universal fires on great-grandchildren because $\text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$. Moving the isolation to a local Corner marker and using the non-transitive piece $\text{comp}(\text{TPPI}, \text{TPPI}) = \{\text{TPPI}, \text{NTPPI}\}$ eliminates that specific pathway—the universal at the grandparent no longer reaches deeper nodes. But a *different* obstruction takes over: the composition table permits any of $\{\text{PO}, \text{EQ}, \text{TPP}, \text{NTPP}, \text{TPPI}, \text{NTPPI}\}$ between two corners at different intended grid coordinates, and nothing in the concept language rules out EQ. The Corner markers themselves then force the EQ-merge. This is exactly the obstruction Wessel identified in report7.pdf, Figure 11 (cf. §2.3): the composition table does not carry the rigidity needed to keep distinct grid cells distinct.

Remark 3.2. The failure mode is structurally different from the RCC5 one but traces back to the same root cause. Lutz–Wolter’s L_{RCC8} reduction *circumvents* the coincidence obstruction rather than enforcing it: they linearize the plane into a single NTPP-chain (the Marx–Reynolds enumeration) and recover the vertical direction from the enumeration, so no “north-of-east = east-of-north” coincidence

has to be forced; the geometry of $\mathbb{R}^n$ only has to keep that linear encoding faithful [9]. No such rigidity is available in the abstract RCC8 semantics of $\mathcal{ALCI}_{RCC8}$: a pure composition table allows corner points to merge freely, and the TPPI/NTPI distinction—while useful locally for a direct/indirect separation at level 2—cannot be chained to enforce an unbounded $\mathbb{N} \times \mathbb{N}$ grid. This matches the open-problem status of $\mathcal{ALCI}_{RCC8}$ decidability noted in [9] and further discussed in [8].

4. Why Naïve Tableau Blocking Fails

Standard DL tableau calculi achieve termination through *blocking*: if a node x has the same label as an ancestor y , we can reuse y ’s subtree for x , cutting off infinite branches. The key requirement is that the “reuse” preserves all constraints—in tree-shaped models, a blocked node’s interactions are limited to its parent edge, so label equality suffices.

In $\mathcal{ALCI}_{RCC5}$, models are *complete graphs*: every pair of domain elements is related by an RCC5 base relation. A node interacts not just with its parent and children, but with *every other node in the model*. This creates a fundamental obstacle for blocking.

Example 4.1 (Blocking breaks with cross-edges). Consider $C_0 = \exists \text{PPI}.A \sqcap \exists \text{DR}.\neg A \sqcap \exists \text{PO}.\top$. A tableau might generate nodes:

- x_0 : root, satisfies C_0
- x_1 : PPI-child of x_0 , type contains A
- x_2 : DR-witness for x_0 , type contains $\neg A$

Now x_1 and x_2 must be related by some RCC5 relation R , and R must be safe for their types. If x_1 generates a descendant x_3 with the same label as x_1 , naïve blocking would declare x_3 blocked. But x_3 ’s relation to x_2 may differ from x_1 ’s relation to x_2 : via composition, $\text{comp}(\text{inv}(\text{PPI}), \text{DR}) = \text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}$ forces x_1 to be DR-related to x_2 , while x_3 (deeper in the tree) may face a different forced relation. The blocked node’s cross-edge commitments differ from the blocking node’s.

Multiple sophisticated blocking strategies have been tried and failed for $\mathcal{ALCI}_{RCC5}$ [47, 36, 35]:

1. **Profile-cached blocking** (GPT-5.4) [35]: Caches “coherent predecessor blocks” to compare global neighborhoods. Fails because the *color structure changes during unraveling*—when a blocked subtree is replayed, the edge-coloring to nodes outside the subtree is not preserved.
2. **Meet-semilattice replay** (GPT-5.4) [35]: Uses a meet-semilattice structure on labels to ensure robust normalization. Same unraveling gap.
3. **Tri-neighborhood blocking** (Claude) [32]: Matches not just node labels but entire *triangle neighborhoods* (the set of all types of 2-element subsets involving the node). **Termination disproved**: frontier nodes continuously acquire new triangle types as new neighbors are added, so $\text{Tri}(x) \subsetneq \text{Tri}(y)$ for frontier x and interior y —blocking never triggers.
4. **Finite-witness / patchwork-augmentation probes** (Claude) [31, 33, 34]: Probes of a finite-witness conjecture and typed-patchwork augmentations for the contextual tableau, both refuted by explicit counterexamples.

The common obstacle is the **blocking dilemma**: making the blocking condition strong enough for soundness (matching all cross-edge interactions) prevents termination (the cross-edge context keeps growing), and vice versa. A full catalogue of retracted approaches is maintained in [47]; the overall project is summarised at the repository [46].

5. Split-Forest Model Decomposition

The breakthrough insight, due to Wessel, is to decompose $\mathcal{ALCI}_{RCC5}$ models into a *tree layer* and a *cross-edge layer*, where the cross-edge layer is simple enough to handle without full global blocking.

Figure 1 reproduces the original whiteboard sketch from April 7, 2026: PPI chains form trees, DR/PO are sibling cross-edges, and DR propagates rigidly downward along PP-chains.

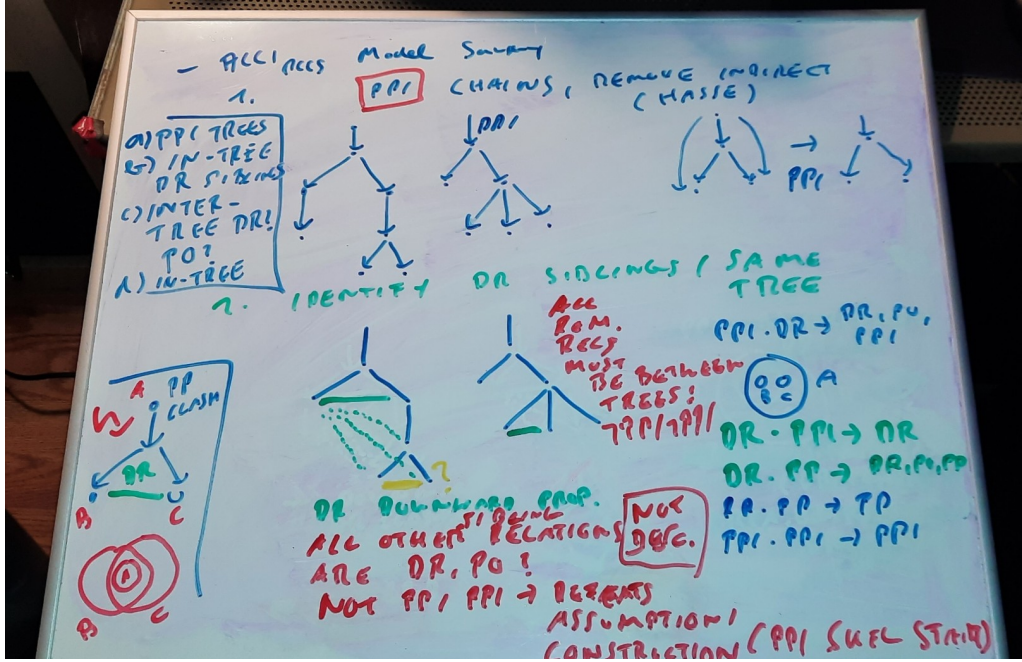


Figure 1: Wessel’s whiteboard sketch (April 7, 2026) of the split-forest decomposition for $\mathcal{ALCT}_{RCC5}$. Vertical PPI-chains form the cover-tree backbone (with join nodes split into EQ-mates); DR/PO become sibling cross-edges between non-comparable subtrees; and DR propagates rigidly downward because $\text{comp}(\text{PP}, \text{DR}) = \text{comp}(\text{DR}, \text{PPI}) = \{\text{DR}\}$, so the residual cross-edge alphabet is just $\{\text{DR}, \text{PO}\}$, which is trivially arc-consistent. Every subsequent formalisation in this paper—GPT-5.4’s split-forest semantics, Claude’s cover-tree tableau, and GPT-5.5’s round-7-through-round-9 sketches—rests on this two-layer decomposition.

5.1. Cover trees

In any $\mathcal{ALCT}_{RCC5}$ model, the PP/PPI relation is a strict partial order (by composition: $\text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$). Orient this order downward (PPI = “has proper part”) to obtain a DAG. If the DAG has join nodes (elements with multiple PP-predecessors), split them into EQ-copies, one per parent path. The result is a rooted forest—the *cover tree*.

Definition 5.1 (Cover tree). A *cover-tree presentation* of an $\mathcal{ALCT}_{RCC5}$ model $\mathcal{I}$ is a rooted tree T with a surjection $\text{orig} : T \rightarrow \Delta^{\mathcal{I}}$ such that: (i) u is the parent of v in T iff $\text{orig}(u)$ PPI $\text{orig}(v)$ (immediate containment), and (ii) tree nodes mapping to the same element are EQ-mates (weak-EQ semantics, convertible to strong-EQ by typed congruence quotient).

5.2. The key observation: DR rigidity

In the cover tree, what relations can hold between nodes in different sibling subtrees? Consider sibling roots s and t (children of the same parent), and elements x below s , y below t in the tree. An element x below s is in the Core of the sibling pair (s, t) if $\text{orig}(x)$ is a common descendant of both $\text{orig}(s)$ and $\text{orig}(t)$ in the original model’s PP order—i.e., $\text{orig}(x)$ is a proper part of both. Core elements are handled by EQ-splitting; the interesting question concerns non-Core elements.

Proposition 5.2 (Wessel). *After EQ-splitting, only DR and PO are possible between non-Core elements of sibling subtrees.*

Proof. • EQ(x, y): In the original DAG (before splitting), $x = y$ would mean a single element is a PP-descendant of both s and t —i.e., it is in the Core. After splitting, this element is duplicated

into distinct tree nodes x' below s and y' below t , with $\text{orig}(x') = \text{orig}(y')$ (EQ-mates in the weak-EQ presentation). Since $x' \neq y'$ as tree nodes, $\text{EQ}(x', y')$ is impossible under strong EQ semantics.

- $\text{PP}(x, y)$: would mean x lies below y in the part-of order, hence below t , making x a common descendant (Core).
- $\text{PPI}(x, y)$: would place y below x below s , meaning t lies below s —contradicting sibling incomparability.

Only DR and PO remain. Furthermore, DR propagates rigidly: $\text{comp}(\text{PP}, \text{DR}) = \{\text{DR}\}$ and $\text{comp}(\text{DR}, \text{PPI}) = \{\text{DR}\}$, so if $s \text{ DR } t$, then *every* descendant of s is DR to t and all its descendants. $\square$

5.3. Trivial arc-consistency of $\{\text{DR}, \text{PO}\}$ networks

The consequence of Proposition 5.2 is that the cross-edge constraint network uses only $\{\text{DR}, \text{PO}\}$ as possible relations (plus $\{\text{DR}\}$ for DR-rigid pairs).

Proposition 5.3. *For all $R, S \in \{\text{DR}, \text{PO}\}$: $\text{comp}(R, S) \cap \{\text{DR}, \text{PO}\} \neq \emptyset$.*

This means a $\{\text{DR}, \text{PO}\}$ disjunctive network is *trivially arc-consistent*: every domain is either $\{\text{DR}\}$, $\{\text{PO}\}$, or $\{\text{DR}, \text{PO}\}$, and composing any two non-empty domains produces a non-empty result. By the patchwork property [2], the network therefore has a globally consistent atomic refinement whenever every domain is non-empty.

This is the central simplification: *checking that each type pair has a non-empty $\{\text{DR}, \text{PO}\}$ -safe set suffices for cross-edge consistency*. No full path-consistency computation is needed.

6. The Cover-Tree Tableau

6.1. Algorithm overview

The cover-tree tableau searches for a *valid type set*: a finite set $T \subseteq \text{Tp}(C_0)$ of Hintikka types that contains a root type realizing C_0 and satisfies four conditions:

- (CT1) **Demand closure**: every existential demand $\exists R.D$ in every $\tau_i \in T$ has an R -safe witness $\tau_j \in T$ with $D \in \tau_j$.
- (CT2) **Cross-edge consistency**: every pair $\tau_i, \tau_j \in T$ has either $\text{Safe}(\tau_i, \tau_j) \cap \{\text{PP}, \text{PPI}\} \neq \emptyset$ (tree-relatable) or $\text{Safe}(\tau_i, \tau_j) \cap \{\text{DR}, \text{PO}\} \neq \emptyset$ (cross-relatable).
- (CT3) **Cover-tree sibling compatibility**: for each type τ_i with multiple DR/PO demands, the witnesses can be jointly assigned such that every witness pair satisfies $\text{comp}(\text{inv}(R_m), R_{m'}) \cap \{\text{DR}, \text{PO}\} \cap \text{Safe}_{\text{DR/PO}}(\tau_{j_m}, \tau_{j_{m'}}) \neq \emptyset$.
- (CT4) **Composition propagation**: for each type with multiple demands of *any* role combination, witnesses τ_{j_1}, τ_{j_2} must satisfy $\text{comp}(\text{inv}(R_1), R_2) \cap \text{Safe}(\tau_{j_1}, \tau_{j_2}) \neq \emptyset$.

The algorithm terminates because $|\text{Tp}(C_0)| \leq 2^{|\text{cl}(C_0)|}$ is finite, and the type set grows monotonically during the search.

6.2. How blocking works

The crucial difference from naïve tableau approaches is the *separation of layers*:

- The **tree layer** (PP/PPI) handles ancestor/descendant demands. Since PP/PPI form a tree, standard tree-model arguments apply: blocking via *rank- d signatures* (the type information visible within modal depth $d = \text{md}(C_0)$) yields a finite representation.

- The **cross layer** (DR/PO) handles demands for disconnected or partially overlapping regions. Because the cross-edge network uses only $\{DR, PO\}$ —which is trivially arc-consistent—the cross-layer check reduces to a simple pairwise non-emptiness condition (CT2), not a global consistency computation.

Naïve blocking fails because it tries to handle both layers simultaneously: the tree structure provides natural blocking, but the cross-edges to distant nodes keep changing, preventing a match. The cover-tree tableau sidesteps this by *not* blocking on cross-edge context—instead, it verifies cross-edge consistency as a separate, global check on the finite type set.

Example 6.1 (A concept that defeats naïve blocking but works with cover trees). Consider $C_0 = \exists PO.A \sqcap \exists PP.\neg B \sqcap \forall DR.A$. In a naïve tableau, the PP-child needs further expansion; blocking requires matching both the label *and* the triangle neighborhood (relations to the PO-witness and all other nodes). Since the PO-witness forces DR-relations to the PP-child’s subtree (via $\text{comp}(PO, PP) = \{DR, PO, PP\}$), the cross-edge context keeps evolving.

In the cover-tree tableau: the PP-witness is an ancestor in the tree, the PO-witness is a cross-edge in a sibling subtree. The $\forall DR.A$ fires only on DR-related nodes—the ancestor is PP-related (not DR), so no conflict. The type set T needs a type for the root (containing C_0), one for the PO-witness (containing A), and one for the PP-witness (containing $\neg B$). Cross-edge consistency (CT2) checks that every pair has a non-empty safe set in $\{DR, PO\}$ —a one-time pairwise check, not an evolving context.

Example 6.2 (Implicit PP via composition forcing). The concept

$$X = \exists DR.\exists PO.C \sqcap \forall DR.\neg C \sqcap \forall PO.\neg C \sqcap \forall PP.X$$

contains no $\exists PP$ demand, yet it requires an infinite model with forced PP edges.

A node r satisfying X has a DR-neighbor a (with $\exists PO.C$) and a ’s PO-neighbor b (with C). By Table 1, the relation between r and b lies in $\text{comp}(DR, PO) = \{DR, PO, PP\}$. But r has $\forall DR.\neg C$ and $\forall PO.\neg C$, and $b \in C$ —so DR and PO are blocked, leaving *only* PP. The PP edge is **forced** by composition plus universal filtering, with no explicit $\exists PP$ in the concept.

Since r PP b , node b is an *ancestor* of r in the cover tree. The recursive $\forall PP.X$ forces b to also satisfy X , creating its own $DR \rightarrow PO$ chain and forcing yet another PP-ancestor above it:

$$r \text{ PP } b \text{ PP } b' \text{ PP } b'' \text{ PP } \dots$$

The model is necessarily infinite. The cover-tree tableau handles this finitely: the type for C -elements can serve as its own PP-ancestor (different domain elements, same abstract state), so a three-element type set $\{\tau_0(C, X), \tau_1(\neg C, \exists PO.C), \tau_2(\neg C, \text{root})\}$ represents the infinite model.

This pattern—PP edges arising purely from composition and universal filtering—is a distinctive feature of $\mathcal{ALCT}_{RCC5}$ and illustrates why the complete-graph semantics require careful handling of implicit spatial constraints.

Example 6.3 (The “PO-loop”: 2-element SAT via symmetric cycles). Naive reasoning suggests

$$Y = C \sqcap \exists PO.\exists PO.C \sqcap \forall PO.\neg C \sqcap \forall DR.\neg C \sqcap \forall PP.\neg C \sqcap \forall PPI.\neg C$$

should be unsatisfiable: any witness chain $d \text{ PO } a \text{ PO } b$ with $b \in C$ needs $\rho(d, b) \in \text{comp}(PO, PO) = \{DR, PO, PP, PPI, EQ\}$, and the four universals $\forall R.\neg C$ seem to block every non-EQ option (each forces $b \in \neg C$, yet $b \in C$).

But Y is in fact **satisfiable**, via the two-element model $\{d, a\}$ with $C^{\mathcal{I}} = \{d\}$ and $\rho(d, a) = PO$. Because PO is symmetric ($PO(d, a) \Rightarrow PO(a, d)$), the chain $d \text{ PO } a \text{ PO } d$ loops back to the root itself: the “third node” of the $\exists PO.\exists PO$ formula is d , under strong EQ identity. Two syntactic positions in the nested existential map to the same semantic element.

The universals $\forall R.\neg C$ constrain d ’s *outgoing* R -neighbours (the sole PO-neighbour $a \in \neg C$ satisfies them), but they do **not** prevent d from *being* a PO-neighbour of a : the witness for a ’s $\exists PO.C$ is d itself,

and a has no $\forall\text{PO}$ -constraint forbidding C -neighbours. This asymmetry between “constraints on my neighbours” and “constraints on what I may be a neighbour of” is peculiar to symmetric roles (PO, DR). The cover-tree tableau correctly reports Y as satisfiable.

Example 6.4 (“Clash-out to EQ”: genuine UNSAT via strong EQ). The complementary pattern

$$Z = C \sqcap \exists\text{PO}.\exists\text{PO}.\neg C \sqcap \forall\text{PO}.C \sqcap \forall\text{DR}.C \sqcap \forall\text{PP}.C \sqcap \forall\text{PPI}.C$$

is unsatisfiable, and the reason is instructive. The chain $d \text{ PO } a \text{ PO } c$ forces $a \in C$ (via $\forall\text{PO}.C$ at d) and $c \in \neg C$ (via $\exists\text{PO}.\neg C$ at a). The relation $\rho(d, c)$ must lie in $\text{comp}(\text{PO}, \text{PO}) = \{\text{DR}, \text{PO}, \text{PP}, \text{PPI}, \text{EQ}\}$. The four universals $\forall R.C$ at d (for $R \in \{\text{DR}, \text{PO}, \text{PP}, \text{PPI}\}$) each require $c \in C$, contradicting $c \in \neg C$, hence *all four non-EQ options are blocked*. Only EQ remains.

Under **strong EQ semantics** (EQ = identity), $\rho(d, c) = \text{EQ}$ forces $d = c$, yielding $d \in C \wedge d \in \neg C$ and a clash. The PO-loop escape of Example 6.3 does not apply here: the endpoint c requires $\neg C$ while d has C , so $d \neq c$ is forced semantically, and the universals then close every non-EQ composition entry.

This example illustrates why *strong* EQ semantics is essential for our undecidability- and decidability-proof machinery. Under weak EQ (equivalence-class) semantics, collapsing d and c would not immediately clash, and one would need additional machinery (equality propagation, congruence-closure) to derive the same conclusion. The strong reading is also the one built into the composition tables used in Renz & Nebel’s patchwork theorem.

Remark 6.5 (Reasoners and tools: a cross-validation suite). This paper references three Python tools used in the companion implementations; before the incompleteness below we fix their roles. The **cover-tree tableau** (`cover_tree_tableau.py`, Section 6) is the main candidate decision procedure—the only tool whose soundness and completeness are claimed for $\mathcal{ALCI}_{\text{RCC5}}$ satisfiability, resting on the patchwork property of RCC5 and the companion completeness-extraction argument. The **quasimodel reasoner** [38] (`alcircc5_reasoner.py`, with cycle-aware wrapper `alcircc5_reasoner_cyclic.py`) is a constructive bottom-up quasimodel search based on disjunctive path-consistency; it is used *only* as a cross-validation oracle for the cover-tree tableau, not as a step of any decidability argument. The **independent model verifier** (`model_verifier.py`) grounds SAT answers in concrete finite RCC5 models, separately checking inverse symmetry, composition consistency on all triples, type-safety, existential witnessing, and recursive concept truth. The three tools together form a cross-validation suite around the cover-tree tableau: cover-tree produces the candidate verdict, the quasimodel reasoner offers an independent opinion, and the model verifier anchors SAT answers in explicit models.

Remark 6.6 (Implementation-level bugs found and fixed in April 2026). Three implementation-level defects were identified in April 2026 and fixed in the cover-tree tableau and the baseline quasimodel reasoner: (i) the baseline quasimodel reasoner was incomplete on cyclic-via-symmetric-role SAT concepts such as the PO-loop of Example 6.3 (closed by an opt-in cycle-close clause in `alcircc5_reasoner_cyclic.py`); (ii) the shared `compute_safe` helper failed to propagate $\forall\text{PP}/\text{PPI}.D$ universals as a whole into successor types (fixed by the standard transitive-role trick); (iii) the cover-tree’s pairwise tree-cross check missed twelve UNSAT patterns of the form $\exists R_1.\exists R_2.A \sqcap \bigwedge_{R \in \text{comp}(R_1, R_2)} \forall R.\neg A$ when intermediate types carry a single demand (fixed by porting the quasimodel reasoner’s role-path compatibility fixpoint as Phase 5 / (CT5)). The decidability argument is unaffected by any of these fixes: all three were soundness gaps in implementations of the candidate decision procedure, not the split-forest / completeness-extraction argument that underpins the theorem. The 911-concept cross-validation continues to match 911/911 before and after every fix; full technical detail (root cause, algebraic justification, regression tests) is in the cover-tree implementation paper [29].

6.3. What makes this “non-standard”

The cover-tree tableau differs from a standard DL tableau calculus in several fundamental ways:

1. **No completion graph.** Standard tableaux build a completion graph node by node, applying expansion rules until saturation or clash. The cover-tree tableau searches over *sets of types* (abstract states), not concrete graph structures. It is closer to a type-elimination or automata-theoretic approach.
2. **No explicit blocking rule.** Standard tableaux have an explicit blocking condition that compares nodes during expansion. The cover-tree tableau achieves finiteness *structurally*: the search space is the powerset of $\text{Tp}(C_0)$, which is already finite.
3. **Global side-checker.** The cross-edge constraints (CT2–CT4) are checked globally over the entire type set, not locally during node expansion. This is what avoids the blocking dilemma: cross-edge consistency is a property of the type set as a whole, not of individual nodes.
4. **Existential demands go to ALL existing types.** In a standard tableau, $\exists R.D$ generates one new R -neighbor. In the cover-tree tableau, the witness can be *any* type already in T —including the root type itself. This reflects the complete-graph semantics: every domain element is related to every other.

6.4. Soundness and completeness

The **primary decidability statement** (as of 29 May 2026) is the GPT-5.5 *split-forest + parity-tree-automata proof* [27] (21 pages): the split-forest *semantic normal form*—witness thinning, generated containment covers, split copies for incomparable superparts, side-interface mosaics, and typed-EQ quotient—together with a reduction of the finite search problem to *non-emptiness of a parity tree automaton* over the finite alphabet of split-forest profiles. The automaton keeps the split-forest model idea (in the proof’s own words, “the automaton does not replace the split forest; it consumes it”) and is used only for the parts most brittle in a hand certificate: regular infinite branches, simultaneous eventuality fulfilment, recursive side contexts, and profile repetition without semantic equality. Parity-tree-automaton non-emptiness is decidable by construction, so the finite-checkability and eventuality obligations come for free. This proof was promoted as a strategic move, not a proof upgrade, and was *then cold-reviewed* in turn (the 5th cold review): it was found to share the no-automata route’s finite-representation keystone gap and was repaired (Theorem B, finite abstraction adequacy, being the load-bearing keystone). That keystone was *then cold-reviewed in turn* (the 6th cold review, by a fresh Opus 4.8 of a different model lineage from the proof’s author) and found unproved—its interface-transformer coherence was asserted, not proved—prompting GPT-5.5’s round-12 repair, which deletes the transformer layer and rests global RCC5 consistency on the *patchwork property* of RCC5, decided by a two-way alternating parity automaton. Round-12 has not itself been cold-reviewed. The overall status remains *strongly supported, not certified*.

The *no-automata* route—which encoded those same brittle parts by a finite certificate with request-closed blocking cycles—was the project’s primary statement through round 10 and is now its historical predecessor (§7). Its most developed manuscript is the round-9 forced-verticalization proof [23] with companion [24]; a cold review of round-9 then found a dual descendant-tower defect and two foundational gaps (finite checkability; cross-label determinacy), and round-10’s repair reduced them to *automaton-shaped* lemmas—which is exactly why the automaton was promoted. The remainder of this subsection develops the shared split-forest normal form and the (now-historical) no-automata certificate machinery, since the automata proof consumes the former and the latter records what the automaton replaces.

The no-automata thread’s final state (D-1 and the round-9 repair, historical). The completeness gap (D-1) that a cold Claude Opus 4.8 review found in round-8 was closed at the manuscript level in round-9. The gap was this: round-8’s cross side/tree exhaustion lemma closes the side/tree gap only on the *finite, depth-bounded* boundary $B_k(u)$, whereas a DR/PO side witness can be RCC5-composition forced to be a proper part of an *infinite* pumped containment tower. The concrete satisfiable witness is

$C'_0 = (\exists \text{PP}.G) \sqcap (\exists \text{PO}.\neg X)$ with $G = (\forall \text{PO}.X) \sqcap (\exists \text{PP}.G)$: the tower $a_1 \text{PP} a_2 \text{PP} \dots$ each carries $\forall \text{PO}.X$, and the root's PO-witness $w \in \neg X$ is forced ($\text{comp}(\text{PPI}, \text{PO}) = \{\text{PO}, \text{PPI}\}$ with PO killed by $\forall \text{PO}.X$, then $\text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$) to be a proper part of *every* tower node, a label the bounded saturated side context cannot carry (machine-checked). In round-9 GPT-5.5 ratified the defect and integrated the cold reviewer's proposed repair, *composition-forced verticalization*: a side witness whose ancestor trace has a PPI-tail is spliced into the cover relation $E_\uparrow$ at a bounded threshold, so equality-aware vertical reachability carries the forced label up the tail and the u - w edge stays the inherited residual PO pair. Three lemmas support the repair — monotone ancestor traces ($\text{DR}^* \text{PO}^* \text{PPI}^*$, PPI absorbing), a bounded threshold after regular replacement, and a global residual frontier in which every remaining pair is DR or PO. We state the result honestly as **strongly supported but not, as written, mechanically certified**: the soundness direction is sound, the architecture is sound, the D-1 gap is repaired at the manuscript level—though round-9 was then itself cold-reviewed (§7.10), which found a dual defect and two foundational gaps and led to round-10 and the switch to the automata route as the primary statement. GPT-5.5 names the bounded-threshold replacement lemma, the typed-equality quotient lemma, and the support-closed mosaic patchwork theorem as the parts most worth future formalization. The remainder of this subsection states the construction at the level the manuscript develops it.

Soundness (valid certificate $\Rightarrow$ strong abstract RCC5 model satisfying C_0). A finite split-forest certificate $\mathcal{Q}$ is unfolded to an occurrence structure $\mathcal{U}(\mathcal{Q})$ in which repeated cycle profiles generate fresh concrete occurrences. The quotient $\mathcal{U}(\mathcal{Q})/\approx$ by the explicit equality-port congruence is a strong abstract RCC5 frame: strong-EQ holds because the quotient identifies exactly $\approx$ -classes; inverse symmetry and RCC5 composition hold because the round-7 label function and triple-composition clause (V6) are verified on the finite occurrence-sensitive certificate. A truth lemma over $\text{cl}(C_0)$ shows that $\mathcal{I}_{\mathcal{Q}} \models C_0$ at the initial occurrence. The full chain rests on GPT-5.4's v1 sibling-interface paper [10] (Thms 1.17–1.19) with the v2 delta [11], plus the round-7 architectural specifics.

Completeness (satisfiable $C_0 \Rightarrow$ valid finite certificate) — *the direction that carried the D-1 gap, repaired in round-9*. A satisfying interpretation is thinned to a witness-generated model via the witness-thinning lemma; the generated containment split-cover construction supplies a weak split presentation with explicit equality mates and side mosaics. A finite-index blocking lemma (without automata) regularises every infinite vertical ray to a finite prefix plus a request-closed cycle word. The certificate profiles are the resulting complete boundary profiles; pair shapes and triple shapes are extracted from the occurrence-sensitive incidences of the regular presentation, not merely from abstract position symbols. Round-8 filed every DR/PO witness in a bounded side context, which cannot record the labels forced by composition with an unbounded ancestor tower (D-1); round-9 tests each selected side witness for composition-forced verticalization *before* ordinary side-context attachment, splicing those with a PPI-tail into the vertical backbone at a bounded threshold so the unbounded tail is carried by equality-aware reachability (§7).

The round-7 architectural move. The central ingredient of round 7 is that finite labels live on *occurrence-sensitive pair shapes*

$$\alpha(u, v) = (s_u, p_u, s_v, p_v, \iota(u, v)),$$

where $\iota(u, v) \in \{\text{self}, \text{eq}, \text{up}, \text{down}, \text{side}_R, \text{front}_R\}$ is an *incidence tag* that distinguishes literal occurrence identity ($u = v$, tag self) from explicit equality-port pairs (tag eq), vertical successor / predecessor pairs (tags up, down), side-mosaic pairs, and residual sibling-frontier pairs. The label function is

$$\ell_{\mathcal{Q}} : \text{Pair}_{\mathcal{Q}} \rightarrow \text{B}, \quad \ell_{\mathcal{Q}}(\text{self}) = \ell_{\mathcal{Q}}(\text{eq}) = \text{EQ}, \quad \ell_{\mathcal{Q}}(\text{up}) = \text{PP}, \quad \ell_{\mathcal{Q}}(\text{down}) = \text{PPI}, \quad \ell_{\mathcal{Q}}(\text{side}_R) = \ell_{\mathcal{Q}}(\text{front}_R) = R.$$

Distinct laps of a blocking cycle have $\iota \neq \text{self}$ (and, absent an explicit equality-port link, $\iota \neq \text{eq}$), so $\ell_{\mathcal{Q}}(\alpha(u_i, u_j)) \neq \text{EQ}$ for any distinct laps: the canonical one-state certificate for $\exists \text{PP}.\top \sqcap \forall \text{PP}.\exists \text{PP}.\top$

is realised with $\iota = \text{up}$ and label PP between successive laps, as intended. Triple composition is verified on a finite triple-shape set Tri_Q via

$$\ell_Q(\alpha_{13}) \in \text{comp}(\ell_Q(\alpha_{12}), \ell_Q(\alpha_{23})) \quad \text{for all } (\alpha_{12}, \alpha_{23}, \alpha_{13}) \in \text{Tri}_Q,$$

the round-7 (V6) clause.

Twelve validity conditions, finite enumeration. All of round 7 is twelve finite-checkable validity conditions (V1)–(V12) on a finite occurrence-sensitive certificate. The certificate space is bounded by a coarse $B(C_0) \leq 2^{2^{p(n)}}$ for a polynomial p , sufficient for decidability. Three structural ingredients carry the proof:

- (1) **Occurrence-sensitive equality.** Equality is generated only by explicit equality-port chains instantiated inside a context, never by reuse of an abstract position symbol in a blocking cycle.
- (2) **Split copies for multiple selected proper superparts.** Where an element has incomparable selected PP-superparts, the presentation carries one equality-mate occurrence per selected superpart.
- (3) **Request-closed cycles** in place of ω -acceptance. Transitive PP/PPI eventualities are discharged by finite-graph reachability plus a request-closure condition: no parity or Büchi automaton.

Computational verification of the combinatorial claims. Three self-contained pure-stdlib scripts exercise the central round-7 stress concepts: WP7 (comparable side witnesses, $\exists \text{DR}.(A \sqcap \exists \text{PP}.B) \sqcap \exists \text{DR}.B$), WP8 (blocking-not-equality: a finite truncation of the canonical one-state $C_\uparrow$ chain verifies that the round-7 incidence-tagged labelling rule keeps distinct laps at PP/PPI rather than EQ, and explicitly demonstrates the discarded round-6 collapsed rule), and WP9 (split copies for $\exists \text{PP}.A \sqcap \exists \text{PP}.B$ with incomparable proper superparts). All three pass. Six earlier work packages (WP1–WP6), written against the round-2 31-page repaired proof [16], also all PASS; their detailed description is in §7 below.

A note on scope. The soundness and completeness statements are theorems about the *split-forest semantics*, not about the implementation in §6. The cover-tree tableau is an algorithmic *realization* of this theory; a formal derivation showing that CT1–CT5 exactly capture the round-7 (V1)–(V12) is a separate claim, empirically supported but not yet written. The companion implementation paper [29] discusses this theory-vs-implementation split in detail.

6.5. Computational evidence

The algorithm is implemented in approximately 350 lines of Python [29, 37]. Cross-validation against an independent quasimodel-based reasoner [38, 39] on **911 test concepts** (50 known SAT, 13 known UNSAT, 36 adversarial, 512 systematic triples, 200 random depth-2, 100 random depth-3) yields **zero mismatches** [42, 43, 44]. A decomposition test [41] confirms that **775/775 SAT concepts (100%)** have finite cover-tree models, with 12.2 M total models enumerated and **zero genuine counterexamples**. Models produced by the tableau are independently replayed by a standalone verifier [40].

Additionally, a GIS taxonomy computation [45] applies the cover-tree tableau to a realistic $\mathcal{ALCT}_{\text{RCC5}}$ ontology from Wessel’s original report [8]: 18 spatio-thematic concepts (countries, cities, rivers, lakes) connected by RCC5 spatial relations. The tableau reproduces **all 21/21 expected subsumptions**, including non-trivial inferences like Hamburg $\sqsubseteq$ GermanCity that require RCC5 composition reasoning through spatial role chains. Figure 2 shows the original computed taxonomy from Wessel’s report [8].

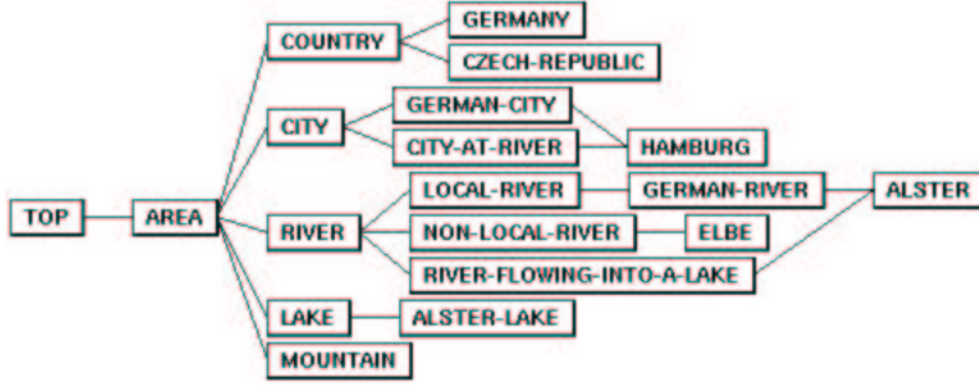


Figure 2: Computed taxonomy of the GIS example TBox from Wessel [8], Figure 6. The cover-tree tableau reproduces all 21 subsumption edges, including non-trivial spatial inferences such as $\text{Hamburg} \sqsubseteq \text{GermanCity}$ (requiring RCC5 composition propagation through the PO-chain Hamburg–Alster–AlsterLake).

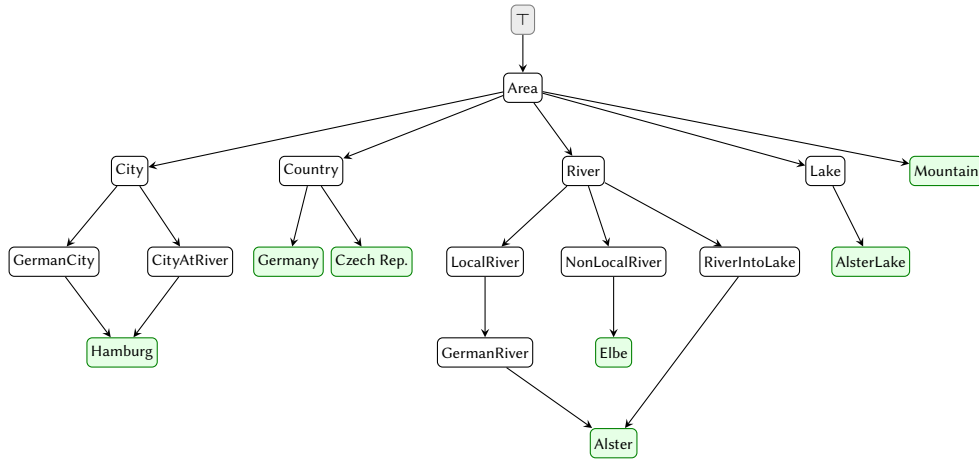


Figure 3: Taxonomy of the GIS example TBox, computed by the cover-tree tableau [45]. Green nodes are leaf concepts (most specific). Note the DAG structure: *Alster* has two parents (*GermanRiver* and *RiverIntoLake*) and *Hamburg* has two parents (*GermanCity* and *CityAtRiver*). All 21 subsumption edges match Figure 2.

7. History of Attempts (April–July 2026)

The decidability argument summarised in §6.4 reached its current form through seven rounds of adversarial review between Claude (Anthropic) and GPT-5.5 (OpenAI) in April–May 2026, prompted by Michael Wessel. This section gives a compact abstracted narrative of the evolution. The full prose of each round, the original superseded manuscripts, and the per-round repair list are preserved in `OUTDATED.md` [47].

7.1. The starting point: cover-tree tableau + v1 completeness extraction (April 2026)

The cover-tree tableau [12, 29] and its companion split-forest semantics [10] were established in April 2026 by GPT-5.4 Pro and Claude Opus 4.7, prompted by Wessel. The split-forest paper proves the soundness direction in full detail (Thms 1.17–1.19); the converse—extracting a finite quotient from a satisfying model—was given only as a proof sketch. Claude wrote a first attempt at the missing completeness extraction (*v1* [14], April 2026). In parallel, the cover-tree implementation underwent two rounds of adversarial review by Claude Opus 4.7 (April 17–18, 2026): twelve implementation-level counterexamples were identified across the 4×4 grid of (R_1, R_2) three-type chains and the PP/PPI-transitive universal pattern, and all were fixed in the shared `compute_safe` helper and the

new Phase 5 / (CT5) role-path compatibility check (Remark 6.6 above). After these fixes the 911-concept cross-validation against the cycle-aware quasimodel reasoner stabilized at 911/911 matches with zero mismatches.

7.2. v1 superseded, v2.1, and the round-2 31-page repaired proof (May 6–12, 2026)

GPT-5.5 reviewed v1 and identified two structural defects: the Hasse construction for the cover-tree broke on infinite PP-chains (canonical example $C_{\uparrow} \equiv \exists PP.\top \sqcap \forall PP.\exists PP.\top$), and the $C5$ triple-coherence proof conflated witnesses across pairs. Claude responded with v2 [13] (May 6) and v2.1 (May 6). A second-round review of v2.1 then identified eight structural gaps, three load-bearing: profile-port equality contradicted vertical separation on self-loops via the $M1+M7$ reflexivity chain; the mate construction forced all mates to share a single semantic parent (refuted by the two-mate stress concept $C_{AB} \equiv A \sqcap B \sqcap \exists PP.A \sqcap \exists PP.B \sqcap \forall PP.(A \rightarrow \neg B) \sqcap \forall PP.(B \rightarrow \neg A)$); and the rootless orientation was incompatible with the depth-from- u_* projection. GPT-5.5 produced a *repaired proof* [16] (May 12, 16 pages) fixing all three defects via occurrence-level equality, mates with different parents, and request-closed cycles in place of ω -acceptance. v2.1 was marked superseded; the repaired proof was adopted as the current best statement. Claude’s contribution shifted to verification: six Python work packages (WP1–WP6) cross-checking the repaired proof’s finite combinatorial content, all PASS.

7.3. Rounds 3 and 4: structural layers added (May 13–21, 2026)

Rounds 3 and 4 of GPT-5.5 review added four further structural layers to the repaired proof, growing it from 16 to 31 pages: (i) a verticalization discipline (mosaic axiom (M6)) stratifying each context into a vertical stratum and a sibling frontier; (ii) port-indexed mate clusters $\text{Mate}_Q(C, s, p)$ attaching mate equivalences to contexts and ports rather than to bare positions; (iii) an explicit abstract triple graph T_Q with cardinality bounds; (iv) a side interface $\text{Side}(u)$ with polynomial width bound $m(n) = O(n^3)$. The certificate-space bound $B(C_0) = 2^{2^{p(n)}}$ became *grammar-first*, derived from these component bounds.

7.4. Round 5: (M6′) recursive verticalization (May 22–23, 2026)

The round-5 review (GPT-5.5, May 22) identified a load-bearing mathematical defect: the (M6) verticalization discipline was too strong. It forbade non-vertical PP/PPI relations between sibling-frontier mosaic positions, but such relations occur in satisfiable witness-generated models—the round-5 stress concept $\exists DR.(A \sqcap \exists PP.B) \sqcap \exists DR.B$ is SAT in abstract RCC5 yet rejected by (M6). Eight repairs were recommended. Claude drafted two follow-up papers (the (M6′) recursive-verticalization delta and the items-4-to-8 delta) that closed all eight items at the paper level, then consolidated them into a 36-page round-5 v2 paper. Side-witness positions were allowed to open nested local vertical strata, with the round-2 (M6) recovered as the depth-1 case.

7.5. Round 6: internal-coherence pass (May 26, 2026)

The round-6 review of the consolidated v2 (GPT-5.5, May 26) found six internal-consistency gaps: the old side-interface lemma still asserted flat $\{\text{DR}, \text{PO}\}$ sibling frontiers; the weakened per-position support closure (SC4′) was inconsistent with the patchwork proof’s per-triple appeal; overlap universal-propagation was under-proved; the lasso construction used the complete profile rather than the extended profile; the side-width bound and validity-clause numbering were inconsistent; and the verification script for the round-5 stress concept was not self-contained. Claude drafted a 39-page round-6 internal-coherence pass closing all six gaps and adding a self-contained pure-stdlib verification script (`wp7_selfcontained_side_witness.py`).

7.6. Round 7: the abstract-position-symbol collapse (May 27–28, 2026)

The round-7 review of the round-6 paper (GPT-5.5, May 27) identified one central architectural defect that could not be locally patched. Round-6 had defined the concrete pair label by $\text{lab}(u, v) := L_Q(q(u), q(v))$, where $q : \Omega(\mathcal{U}(Q)) \rightarrow \text{Pos}_Q$ sends each occurrence to an abstract position symbol and L_Q is the abstract pair-label function with standard reflexivity $L_Q(\pi, \pi) = \text{EQ}$. In a finite regular certificate the abstract position alphabet is finite while the unfolding is infinite, so blocking deliberately reuses the same symbol on fresh occurrences. For the canonical one-state upward cycle, distinct laps u_i, u_j share the same symbol π and the round-6 rule gives $\text{lab}(u_i, u_j) = L_Q(\pi, \pi) = \text{EQ}$, even for $i \neq j$: the strict PP chain collapses into a PP-self-loop. The defect is intrinsic to indexing labels by position-symbol pairs.

GPT-5.5’s round-7 fix [18] replaces the position-symbol pair labelling by labels on *occurrence-sensitive pair shapes* with explicit incidence tags (described in §6.4). Twelve validity clauses (V1)–(V12) replace round-5 / round-6’s fifteen. Claude wrote a round-7 audit/response [28] mapping the round-6 (G1)–(G6) and round-7 (D1)–(D5) defect lists to their resolutions in the round-7 architecture, plus three further self-contained verification scripts (WP7–WP9). All of Claude’s round-5 / round-6 manuscripts are superseded by round 7 and retained as the historical audit trail in `papers/gpt5.5_round2/` [47].

7.7. Round 8: saturated side contexts and constructed catalogues (May 28, 2026, evening)

A fresh Claude Opus 4.7 session, given the round-7 sketch with no prior project context, identified three architectural defects. First, the round-7 pair-shape enumeration did not exhaustively cover the case of a side witness paired with a vertical-boundary position: a PP or PPI label between an exposed boundary point and a side position could be hidden in the residual DR/PO frontier closure. Second, the residual DR/PO frontier restriction was imposed *before* the side context was saturated, making the restriction unsound while PP/PPI pairs were still pending. Third, the pair and triple catalogues Pair_Q and Tri_Q were asserted finite but not explicitly constructed.

GPT-5.5’s round-8 repair [19] closes all three by introducing a *saturated side context* $S_k(u)$ generated by five (Sat1)–(Sat5) operations: start with the exposed vertical boundary of the source, add one selected witness for each local $\exists R.D$ with $R \in \{\text{DR}, \text{PO}\}$ and modal depth strictly less than k , recursively add smaller-depth witnesses, assign or inherit a complete RCC5 label to every ordered pair (including side/tree pairs), and close under equality (label EQ triggers an equality-port declaration) and containment (label PP or PPI triggers a local vertical stratum) *before* applying the residual DR/PO restriction. A new *cross side/tree exhaustion lemma* formalises that no PP/PPI side/tree pair is hidden in the residual frontier. Pair_Q and Tri_Q are now constructed as the least finite sets containing the explicit families of pairs and triples generated by the saturated side contexts, overlap amalgams, and residual configurations, with every triple required to satisfy the RCC5 composition table. The 15-page round-8 sketch and the 30-page detailed companion [20] are filed under `papers/claude_latest_review_gpt5.5_fix/`. Round-8 was the most developed statement until a cold Opus 4.8 review found the D-1 completeness gap (next); round-9 is its successor.

7.8. Round 8 cold review: an open completeness gap (D-1) and a proposed repair

A further cold review — this time by Claude Opus 4.8, again with no prior project context — examined whether the saturated-side-context repair actually closes the side/tree gap, and found that it does not: it relocates the gap rather than closing it [21].

The cross side/tree exhaustion lemma quantifies only over the *finite, depth-bounded* boundary $B_k(u)$ and the finite saturated side context $S_k(u)$; both are bounded by the side-width $m_*(C_0)$. The lemma is true as stated, but the advertised global conclusion — “no PP/PPI side/tree pair is hidden” — is a statement about the whole unfolded model, in which the vertical ancestor chain is infinite. The genuinely hard case is a DR/PO side witness that RCC5 composition forces to be a proper part of an

infinite pumped tower. The concrete satisfiable witness is

$$G := (\forall \text{PO}.X) \sqcap (\exists \text{PP}.G), \quad C'_0 := (\exists \text{PP}.G) \sqcap (\exists \text{PO}.\neg X).$$

G generates an infinite ascending tower $a_1 \text{PP} a_2 \text{PP} \dots$, each $a_m \in \forall \text{PO}.X$; the root's PO-witness $w \in \neg X$ is forced to satisfy $\rho(a_m, w) = \text{PPI}$ for *all* m (base case: $\rho(a_1, w) \in \text{comp}(\text{PPI}, \text{PO}) = \{\text{PO}, \text{PPI}\}$, and $\forall \text{PO}.X$ at a_1 with $w \in \neg X$ kills the PO option; induction: $\text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$). Only finitely many tower nodes fit in $S_k(u)$; for the unbounded tail the constructed pair-shape catalogue offers no admissible shape (side_{PPPI} needs co-residence in one bounded context; down needs a cover-edge path the construction never creates for a side witness; front_{DR} is composition-illegal; front_{PO} violates the $\forall \text{PO}.X$ universal-safety check). So the round-8 completeness construction produces an invalid certificate for the satisfiable C'_0 – a genuine gap (“D-1”), of medium-high severity. The composition arithmetic is machine-checked.

The decidability *theorem* most likely survives: placing w as a true vertical descendant of the tower (a selected $E_{\uparrow}$ -child of a_1 , with the u - w edge demoted to a residual PO frontier) appears to yield a valid certificate, and the decision procedure accepts whenever *some* valid certificate exists. The accompanying repair note [22] turns this observation into a construction – *composition-forced verticalization*: detect when composition forces a DR/PO witness to be a proper part of an ancestor, and splice an $E_{\uparrow}$ cover edge at the (provably bounded) threshold height where it first becomes a proper part, so equality-aware reachability carries the correct PP/PPI label up the entire tail. Three facts make the repair clean, the first machine-checked: (i) the forced label up an ascending chain is monotone in $\text{DR} \succ \text{PO} \succ \text{PPI}$ and PPI-absorbing, hence eventually constant at a threshold; (ii) the threshold is bounded by the number of boundary descriptors, so finiteness is preserved; (iii) the construction *presents a fixed consistent model*, so RCC5 composition and universal safety are inherited rather than re-derived. The cold reviewer recorded this as a candidate with open bookkeeping obligations; GPT-5.5 then took it up in round-9 (next).

7.9. Round 9: composition-forced verticalization closes D-1 (May 29, 2026)

GPT-5.5 ratified the D-1 defect and integrated the cold reviewer's repair into a round-9 proof [23] (16-page sketch) with detailed companion [24] (33 pages), filed under papers/gpt5.5_round3/. A 5-page formal response memo [25] records that GPT-5.5 re-ran the three Opus 4.8 verification scripts (all pass), confirmed the composition facts $\text{comp}(\text{PPI}, \text{PO}) = \{\text{PO}, \text{PPI}\}$ and $\text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$ that drive the phenomenon, and agreed that “this is a genuine completeness defect, not merely a presentational ambiguity.”

Round-9 distinguishes two cases for a selected DR/PO side witness w of a source u . If its relation to the ancestor tower remains DR/PO, w stays an ordinary side witness and deep ancestor-witness pairs are residual DR/PO pairs. If RCC5 composition forces w to be a proper part of all sufficiently high tower nodes, the proof inserts a generated cover edge $w' E_{\uparrow} a^*$, where a^* is the first retained representative of the eventual PPI-tail and w' is w or a split equality mate; every later pair (a_m, w') is then presented by equality-aware vertical reachability. This is a presentation change, not a semantic change: the splice is added only when the inherited satisfying model already has $w' \text{PP} a^*$, and the u - w relation stays the inherited DR/PO residual pair.

Three lemmas carry the repair. *Monotone ancestor traces*: for a DR/PO witness w of u and a tower $u < a_1 < a_2 < \dots$, the trace $r_m = L(a_m, w)$ has shape $\text{DR}^* \text{PO}^* \text{PPI}^*$ (EQ, PP impossible; PPI absorbing because $\text{comp}(\text{PPI}, \cdot)$ never increases rank). *Bounded threshold after regular replacement*: augmenting the tower descriptor with the three-valued trace status leaves finitely many descriptors, so each of the at most three constant regimes needs only boundedly many representatives and the first retained PPI-point sits at a computable height $N_{\text{thr}}(C_0)$. *Global residual frontier*: after equality ports, finite containment saturation, and forced verticalization, every remaining residual-frontier pair has label DR or PO. The split-cover and extraction proofs are updated to apply the forced-verticalization test before ordinary side-context attachment.

The repair preserves the no-automata split-forest architecture (no Büchi/parity automata); the only extra finite information is the three-valued ancestor-trace status, absorbed into the existing finite-index replacement argument. On the computational side, the round-9 construction is exercised end-to-end by a certificate-emission test on C'_0 and its UNSAT sibling, generalised to a parameterised forced-tower family and cross-checked against a by-construction oracle and the cover-tree tableau (§7.12); this closes the round-8 review’s recommendation to add C'_0 as a constructive emission test. GPT-5.5 explicitly characterises round-9 as “a serious proof manuscript rather than a mechanically certified theorem,” naming the bounded-threshold replacement lemma, the typed-equality quotient lemma, and the support-closed mosaic patchwork theorem as the parts most worth future formalization. When written, round-9 was *strongly supported but not certified* and the most developed statement of the no-automata thread. It was then cold-reviewed in turn—which found a dual descendant-tower defect and two foundational gaps—and superseded within the day, as the next subsection records.

7.10. Round 9 cold review, round 10, and the switch to the automata route (May 29, 2026)

A fresh GPT-5.5 session, given the round-9 manuscripts with no project context, reviewed round-9 and found it was not yet a proof. The composition table was independently re-verified correct. Three *critical* defects:

- **One-sided verticalization.** Round-9 handled a DR/PO witness forced into an infinite *ancestor* PPI-tower but missed the dual case — a PO witness forced to be a proper *superpart* of an infinite *descendant* PP-tower. Concrete witness $C = (\exists \text{PO}.A) \sqcap (\exists \text{PPI}.\top) \sqcap \forall \text{PPI}.\top \sqcap \forall \text{PPI}.\top \sqcap \forall \text{DR}.\neg A \sqcap \forall \text{PO}.\neg A$: each descendant d_i is forced ($\text{comp}(\text{PP}, \text{PO}) = \{\text{DR}, \text{PO}, \text{PP}\}$, DR/PO killed by the universals, $\text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$ absorbing) to satisfy $d_i \text{PP}w$ (machine-checked).
- **Finite checkability never proved.** The validity clauses (V6)/(V7) were *asserted* to be finite syntactic checks but quantify over the infinite unfolding; the finite pair/triple catalogues were never shown to exhaust it.
- **Boundary non-determinacy.** A boundary descriptor does not determine the mixed outside/internal labels: $i\text{PP}b, o\text{PO}b \Rightarrow L(o, i) \in \{\text{DR}, \text{PO}, \text{PPI}\}$.

plus four majors, including a *false* simple-cycle bound (a request-closed cycle may need a closed walk that revisits descriptors).

Round 10 accepts all findings and repairs them: a dual descendant splice (symmetric to round 9), occurrence-local requests, ranked side-width, global-quotient-before-cycle-selection, a corrected $(M+1)N$ closed-walk bound, and — for the two foundational criticals — a *finite product-state exhaustion* construction and *complete interface descriptors*. A close reading finds that the entire decidability burden then funnels into two named lemmas, Finite Product Exhaustion and Complete-Interface Replacement, both established by case analysis plus finiteness of a descriptor alphabet, and both *automaton-shaped*: product-state exhaustion is product-automaton reachability, and the complete interface descriptor is a finite interface abstraction.

This is the convergence point. Over rounds 2–10 the no-automata route repeatedly rediscovered automaton-shaped obligations (request-closed cycles for ω -acceptance; product-state exhaustion for finite checking) that it could not discharge. The decision (29 May 2026) is therefore to promote the split-forest + parity-tree-automata proof [27] (§6.4) to the primary statement: it keeps the entire split-forest normal form and lets a parity tree automaton — whose non-emptiness is decidable by construction — handle the brittle half directly. The promotion was strategic, not a proof upgrade; the automata proof was then cold-reviewed in turn (the 5th cold review), found to share the finite-representation keystone gap, and repaired; its finite-abstraction keystone (Theorem B) was then cold-reviewed in turn (the 6th cold review) and found unproved, prompting GPT-5.5’s round-12 patchwork-bag repair (global RCC5 consistency via the patchwork property, decided by a two-way parity automaton), which is now the canonical statement and has not itself been cold-reviewed.

7.11. What survives, and why

The split-forest architecture itself—witness thinning, generated containment covers, weak-EQ split copies for multiple selected PP-superparts, sibling interfaces recorded by finite support-closed mosaics with $\{\text{DR}, \text{PO}\}$ -only open cross-relations, typed equality-port congruence quotient to recover strong-EQ semantics, and request-closed cycles—is present in every round from v1 through round 7. What changed was *how labels are indexed on this architecture*. The v1, v2, v2.1, and round-2 versions indexed labels by the rank- d quotient and abstract position symbols; the round-7 architecture indexes them by occurrence-sensitive pair shapes. The whiteboard sketch (Wessel), the DR-rigidity insight, the EQ-splitting and typed-EQ-quotient idea, the cover-tree decomposition, GPT-5.4’s v1 formalisation, and Claude’s cover-tree tableau implementation all remain load-bearing in round 7.

7.12. Computational verification across rounds

In parallel with the proof work, nine Python work packages verify the finite combinatorial content of the proof family. WP1–WP6 verify the round-2 31-page repaired proof [16] and all PASS: brute-force enumeration confirms the RCC5 composition table in all 25 cells (WP1); a validity-condition checker accepts $C_{\uparrow}$ via parent self-loops, accepts C_{AB} via two mates with different parents, and rejects the v2.1-style single-parent C_{AB} attempt (WP2); a small-model SAT/UNSAT oracle exhibits a concrete 3-element model for C_{AB} (WP3); C_{AB} and request-closed cycle tests fold into WP2/WP3 (WP4, WP5); a mosaic enumeration confirms support-closure axioms on small mosaics (WP6). WP7–WP9 verify the round-7 central stress concepts (comparable side witnesses, blocking-not-equality, and split copies for incomparable proper superparts). WP10 drives the D-1 witness C'_0 end-to-end through the round-9 forced-verticalization construction—detect the forced ancestor trace, splice at the bounded threshold, build the incidence-tagged certificate, validate the round-9 clauses—emitting a valid certificate for C'_0 and rejecting its UNSAT sibling; WP11 generalises this to a parameterised forced-tower family (period- p profiles, many side witnesses, universals over $\{\text{PO}, \text{PPI}, \text{DR}, \text{PP}\}$) and cross-checks the round-9 emit/reject verdict against a by-construction oracle, an independent certificate validator, and the cover-tree tableau (216/216 agreement on the current sweep, no completeness or soundness gap found on that family; the overlap-amalgam case remains future work). WP10/WP11 close the round-8 review’s recommendation to add C'_0 as a constructive certificate-emission test. All PASS. No counterexample to any combinatorial claim of any round was found.

7.13. From automata to Lean: how the argument settled

By the end of May 2026 the decidability argument had reached its automata form (Theorems A/B/C, §6.4), and we regarded that as a natural consolidation point. It did not remain one. A materially stronger reviewer became available (Claude Fable 5, a capability tier above the round-12 author), and the adversarial-review series resumed: across July 2026 the work went through a further sequence of reviews and repairs, migrated its load-bearing soundness argument into the Lean 4 proof assistant, and—most importantly—localized the entire remaining difficulty into a single, well-posed lemma. Rather than recount that as a round-by-round chronology (the blow-by-blow ledger lives in the repository’s CONVERSATION.md, and the formalization history in LEAN.md), we describe the *routes* that were investigated and where each one led. This is the honest shape of the project as it now stands.

7.14. The routes investigated

Every route shares one foundation and differs only in how it tries to make a *decision procedure* rigorous on top of it.

The shared foundation: the split-forest normal form (Theorem A). Every satisfiable concept admits a witness-generated, tree-like *split-forest* model—obtained by witness thinning, generated containment covers, split copies for incomparable superparts, side-interface mosaics, and a typed-EQ

quotient. This semantic normal form is the one component that has *never* broken: it was found sound by every cold review across the project. All of the routes below consume it rather than replace it; they differ in how they turn “a model of this shape exists” into “an algorithm can decide whether one exists.”

Route 1 — the cover-tree tableau (operational). Claude’s cover-tree tableau (§6) is a concrete, implemented decision procedure with exact-occurrence blocking. It is trustworthy on satisfiable inputs and has a known blind spot on certain unsatisfiable inputs (the strong-EQ PO-loop pattern). It is the project’s operational oracle and cross-validation tool—not the decidability proof, but the thing every proposed proof was checked against.

Route 2 — the no-automata split-forest certificate (rounds 2–10). A hand-built finite certificate for satisfiability, validated by local checks over the split forest. Across rounds 2–10 it repeatedly reduced to automaton-shaped machinery it could not fully discharge: request-closed cycles behave like Büchi/parity acceptance, and “finite product-state exhaustion” behaves like product-automaton reachability. Successive cold reviews found successive defects—an ancestor-tower gap, then the dual descendant tower, then that finite-checkability was asserted rather than proved. The thread converged, honestly, onto an automaton.

Route 3 — the automata split-forest proof (rounds 11–12). This made the convergence explicit and became the canonical statement for a time: Theorem A (normal form) $\rightarrow$ Theorem B (a *finite abstraction* captures saturated split forests up to satisfiability) $\rightarrow$ Theorem C (a two-way alternating parity tree automaton accepts exactly the valid abstractions, and its emptiness is decidable). Theorem B is the single load-bearing keystone. Cold reviews (the 5th and 6th) found its global-coherence step *assumed, not proved*; round 12 rebuilt the abstraction on *patchwork bags* so that global RCC5 consistency follows from the citable patchwork property of RCC5 (Renz–Nebel), with eventualities decided by two-way automaton emptiness (Vardi). A 7th (hot) review then found a residual Shadow-Amalgamation gap in the truth lemma. The keystone had become smaller and rested on standard external theorems, but it was still the keystone.

Route 4 — the cluster-quasimodel decision layer (rounds 15–18). Following a cold review’s recommendation, the tree automaton was dropped in favour of a mosaic / type-elimination procedure on the infinitary abstraction directly: a finite *catalogue* of interface patterns carrying *steering functions*, with one-step demand fulfilment (no promissory witnesses, hence no parity/eventuality condition). This is the route on which the whole difficulty was finally *localized*. Once the construction was written explicitly, everything reduced to two named lemmas: **F6**, a computable bound on the *live width* of the certificate (the genuine, non-shadow branching an interface must record), and **W2’**, uniformization of the extraction (that locally realizable glue steps factor through the finite catalogue’s states). Cold reviews sharpened the steering condition from pointwise to joint and pinned down the transport rule; no new mathematical obstacle beyond F6/W2’ appeared.

Route 5 — the Lean formalization (rounds 19–30). The certificate transport, S-layer, catalogue, and logic layers were ported into Lean 4 (core, no `mathlib`). The result is a kernel-checked development (`formal/Round19Transport.lean`, zero sorry) in which the *soundness* pipeline—certificate syntax $\rightarrow$ a genuine RCC5 model of the logic—and the *faithfulness* of the Hintikka abstraction (the type-level certificate loses nothing relative to models) are proved by the kernel. On top of it sits a *decision-grade, non-oracular* reduction: a fixed Boolean checker of first-order finite certificate data, proved sound, from which `Decidable` satisfiability follows by finite search—leaving exactly one open premise, a computable complete enumeration of bounded certificates, which is precisely $F6 \wedge W2'$. Two cold reviews of the Lean confirmed there is no *soundness* defect and found only completeness-interface gaps, which subsequent rounds closed. The concrete history is in `LEAN.md`.

The reduction to one keystone, and the recent attacks. All five routes bottom out at the same place: *does some concept force unbounded live width?* Two independent cold research attacks then sharpened this question. Every naive counterexample—shell half-graphs, prefix pumps, single-chain PO-gaps—collapses to a finite catalogue, so raw pairwise non-mergeability is the wrong invariant; the sharp obstruction is a forced, unbounded family of *identity-selector minors* (private matching of unbounded hitting number). $W2'$ was shown to *fold into* F6 (its coarse form is false, but repairable by a state refinement whose size F6 itself bounds). A regular-cover reframing—finite generators whose infinite unfoldings are the models—gave a complete *local* algebra, of which one durable piece was harvested and machine-certified: the RCC5 *normal form* (every strong-EQ RCC5 network is an ordered-disjoint structure—a strict partial order for PP plus a downward-closed disjointness for DR, with PO residual). But the regular-cover route’s own keystone is F6 relabelled, and its authors concede the remainder “cannot be a purely local RCC5 composition problem.”

The knife’s edge. F6 is not merely the decidability keystone; it is *the same fact* as the obstruction to proving the problem *undecidable*. To force undecidability one would force a rigid coordinate grid, and the abstract composition table cannot force the coincidence condition (“east-of-north = north-of-east”)—exactly because it cannot pin a horizontal relation to a single value. Turn that looseness into a bound and one obtains decidability; circumvent it and force a grid and one obtains undecidability. The two open directions are one question seen from two sides, which is why neither has been settled in over twenty years.

8. Conclusion and Outlook

8.1. What has been achieved

Decidability of $\mathcal{ALCT}_{RCC5}$ remains *open*. What the project has produced, and what is genuinely certified, is the following.

- **The sound semantic foundation.** The split-forest normal form (Theorem A) [10, 11]: every satisfiable concept has a tree-like split-forest model. Found sound by every cold review; never broken.
- **A machine-checked soundness pipeline and abstraction-completeness.** In Lean 4 (zero sorry): a valid finite certificate unfolds to a genuine RCC5 model of the concept, and the Hintikka abstraction is faithful (satisfiability and Hintikka-realizability coincide). The soundness half of the decidability equivalence is thus kernel-certified.
- **A machine-checked, non-oracular decision-grade reduction.** A fixed Boolean checker of first-order finite certificate data, proved sound, yielding `Decidable` satisfiability by finite search modulo a single open premise—which is exactly $F6 \wedge W2'$.
- **The problem reduced to one well-posed keystone.** The entire remaining difficulty is F6 (bounded live width), sharpened to the identity-selector-minor dichotomy, with $W2'$ folded in. This is a single lemma a future solver can aim at, not a diffuse open question.
- **A certified local theory and an unconditionally decidable fragment.** The RCC5 normal form is machine-certified (`formal/RCC5NormalForm.lean`). The $\forall$ PO-free fragment—concepts with no $\forall$ PO subformula, but *retaining* $\exists$ PO, $\forall$ DR, $\exists$ DR, and all vertical part-of modalities, hence still a genuinely expressive spatial fragment—is unconditionally decidable [30]: it is the syntactic case of the two-tier PO-coherent fragment in which PO—the only relation the composition table cannot pin to a single value—is the sole source of difficulty, and is therefore harmless.
- **The undecidability routes shown blocked** (§3), with the coincidence obstruction identified as the fact shared with the decidability keystone.

- **A methodology.** Adversarial cold review as a standard step, and a division of labour between models, both discussed below.

The formalization details (the round-19–30 progression, the two cold reviews of the Lean, the exact theorem names and axiom footprints) are recorded in `LEAN.md`; `README.md` gives the current-status summary.

8.2. Dead ends

Several routes and proof strategies were pursued and abandoned. Each is a genuine negative result about the problem, not merely a failed attempt.

- **A hand certificate for satisfiability.** The no-automata split-forest thread (rounds 2–10) repeatedly reduced to automaton-shaped machinery it could not discharge by hand (request-closed cycles behave like Büchi/parity acceptance; “finite product-state exhaustion” like product-automaton reachability); it converged, honestly, onto an automaton. The lesson—the keystones a hand proof kept rediscovering (finite checkability, simultaneous eventuality fulfilment) are exactly what an automaton discharges by construction—is why the automata route exists.
- **Route-dependent global coherence.** The round-11 interface-transformer abstraction derived a globally consistent labelling of an infinite tree from purely local checks, assuming a confluence law it never proved; a cold review broke it. It was replaced by resting global consistency on the citable RCC5 patchwork property.
- **Forcing a counterexample by raw crowds.** Every naive attempt to force unbounded width—shell half-graphs, prefix pumps, single-chain PO-gaps—collapses to a finite catalogue. Raw pairwise non-mergeability is the wrong invariant; the certificate-relevant one is unbounded row/profile inequivalence surviving bounded refinement (identity-selector minors).
- **Proving F6 by local compression.** An algebra-only “compress the support tree” proof of F6 is refuted: arbitrary jointly-realizable RCC5 glue steps can have unbounded separator cover (machine-checked), so any F6 proof must use that the bad steps are *not* $ALCI_{RCC5}$ -forceable, not merely RCC5-consistent.
- **Ramsey-type combinatorics.** Wrong quantifier order: Ramsey theory gives unavoidable structure in a large fixed object, whereas F6 concerns structure a formula can *force*. The substrate is Ramsey-friendly (large irreducible $\{DR, PO\}$ crowds exist freely), which is exactly why Ramsey does not fit.
- **Coarse uniformization ($W2'$).** Its coarse form is false: two occurrences of identical coarse state can be forced apart by their relation to a third occurrence outside the interface. It is repaired by a bounded state refinement—and that refinement is bounded by F6, so $W2'$ folds into F6 rather than standing beside it.
- **The regular-cover reframing as a shortcut.** A finer, cleaner vocabulary (finite generators, infinite unfoldings) that subsumes the width view and exhausts the local algebra—but its own keystone is F6 relabelled, and it concedes the remainder “cannot be a purely local RCC5 composition problem.” A better map, not a shorter path.

8.3. Missing hard pieces

- **Decidability itself (F6)—the keystone.** Whether some concept forces unbounded live width—equivalently, an unbounded rigid horizontal crowd, equivalently unbounded identity-selector minors. This is the 24-year-old open problem, now cleanly stated; $W2'$ folds into it. It is provably the *same* fact that blocks a proof of undecidability.

- **The converse of the RCC5 normal form.** The forward direction (every RCC5 network is ordered-disjoint) is certified; the converse (ordered-disjoint structures realize RCC5 closure) is exhaustively machine-checked at small size but not yet formalized.
- **A fully wired, human-reviewed decision procedure.** The Lean decision-grade reduction is non-oracular and sound, but its completeness premise is exactly F6; and no part of the development has been reviewed by human domain experts.
- **Exact complexity.** The PO-coherent fragment has a doubly exponential quotient bound (2^{EXPTIME}) [30]; a matching EXPTIME upper bound for it is not established, and no complexity bound is claimed for the full logic.
- $\mathcal{ALCI}_{RCC8}$. RCC8 also has the patchwork property, so the approach should extend; the TPP/NTPP distinction may require a finer status partition. Not worked out.

8.4. Final assessment

After roughly thirty repair rounds and fifteen adversarial reviews, the project has reached a genuine *local optimum*. The local algebra of RCC5 is exhausted and, in one case, certified; the soundness side of the decidability argument is machine-checked; and the entire remaining difficulty has been compressed into a single well-posed lemma (F6) which is provably the *same* fact that blocks a proof of undecidability. Each further round now buys a smaller delta—a cleaner restatement of the same keystone, or a more specific local lemma—rather than movement on the summit. The honest one-line status is unchanged from the project’s standing label: *strongly supported, not certified*, with the soundness pipeline, the abstraction faithfulness, and the RCC5 normal form the certified exceptions.

We therefore offer the work in the shape it has actually taken: not a resolution of a two-decade open problem, but that problem *reduced*—honestly, checkably, and with a machine-checked surround—to its irreducible core, together with a map of why that core is hard and a named toolbox for attacking it. If the question is ever settled, we expect the resolution to begin from a map of this shape.

8.5. On LLM capabilities, strengths, and weaknesses (May 2026)

This subsection records an honest assessment, by Claude, of what was observed about frontier-LLM capabilities across this project. The assessment is grounded in specific evidence from the April–May 2026 review cycle; it is not a statement about LLMs in general. We expect the picture to change quickly.

Division of labour that worked. After several wrong configurations, the team converged on a clear split. **GPT-5.5** owned the proof: it produced the round-2 31-page repaired proof [16], the round-3 and round-4 structural additions, and the round-7 pair-shape / incidence-tag fix [18] that made the labelling rule sound on blocking cycles. GPT-5.4 Pro authored the v1 split-forest semantics [10] that the entire approach rests on. **Claude (Opus 4.7)** owned implementation, verification, audit, and documentation: the cover-tree tableau [29], the cross-validation suite (911 concepts, zero mismatches), the WP1–WP9 verification scripts, and the round-by-round audit trail. **Wessel** owned research direction: the whiteboard sketch (cover-tree intuition, DR-rigidity, EQ-splitting) on which the entire proof rests, the high-level questions, the correction of conceptual errors (the most consequential being catching GPT-5.4’s initial assumption that PP/PPI could remain as cross-edges after splitting), and the judgement of when to push deeper and when to converge.

The two-layer architecture (theoretical proof + executable verification) was not optional. The empirical layer repeatedly caught defects in the theoretical layer that pure self-review would not have surfaced—most notably the round-5 stress concept $\exists \text{DR}.(A \sqcap \exists \text{PP}.B) \sqcap \exists \text{DR}.B$, which was first identified as SAT by Claude’s reasoners and only then explained as a structural defect of the (M6) verticalization discipline in GPT-5.5’s round-5 review.

Where Claude was weaker than GPT-5.5 (on rounds 5–7). (Read with the calibration below: the round-8 episode shows GPT-5.5 shares these weaknesses, so this is not a flat capability verdict.) Concretely on rounds 5–7: (i) Claude did not spot architectural defects in its own drafts. The round-6 critical bug $\text{lab}(u, v) := L_Q(q(u), q(v))$ with $L_Q(\pi, \pi) = \text{EQ}$ collapsing distinct blocking laps into semantic equality was a basic invariant violation of the central blocking-not-equality slogan; GPT-5.5 caught it on review. (ii) Patch-style fixes accumulated complexity. Claude’s response to a named gap was usually to add machinery (recursive verticalization, port-indexed mate clusters, declared cross-label functions, modal-depth recurrences, the validity-clause numbering growing from (V1)–(V10) to (V1)–(V15)). Each patch closed the named gap but introduced new internal-consistency issues. GPT-5.5’s round-7 fix changed the indexing domain instead of adding a layer: five round-5 / round-6 validity clauses collapsed into one (V6) triple-shape clause, and the manuscript dropped from 39 pages (round-6 consolidation) to 14 pages (round-7 sketch). (iii) Claude did not reliably know when to stop iterating.

Are the two models on par? A calibrated read. It is tempting to read the review episodes — Claude finding defects in GPT-5.5’s proofs — as evidence that Claude is the stronger model. On this project’s evidence that conclusion is not warranted; the honest read is *broadly comparable frontier models with the same blind spots*. The comparison is role-asymmetric (GPT-5.5 was assigned proof-author, Claude verification and review; the symmetric experiment was never run) and version-confounded (“GPT-5.5” was one model throughout, while “Claude” was Opus 4.7 for rounds 5–6 and Opus 4.8 for the round-8 review, so the D-1 catch is partly a newer model out-reviewing a months-older collaboration). Moreover, cold review is a different and arguably easier task than authoring: finding D-1 in a finished proof from a cold start is not the same as producing the proof from a blank page, and the reviewer-was-cold property is the load-bearing one (a fresh GPT-5.5 might have found D-1 too — untested). Both models have demonstrated genuine proof-*construction*, not just one of them: GPT-5.5’s round-7 pair-shape fix was the project’s most elegant structural move, Opus 4.8’s D-1 repair was substantive new proof work, and GPT-5.5 then ratified that repair and integrated it into round-9 with its own monotone-trace and bounded-threshold lemmas — construction work on both sides of the author/reviewer line. The robust finding, independent of which model is sharper, is that *neither model is self-sufficient*: both write fluent long-context formal mathematics, both fail to reliably self-review, both declare results “closed” prematurely (GPT-5.5 advertised round-8 as fixing the round-7 gap, and it still had a deeper one; round-9 is now the latest “closed” claim, and by the logic below it should be distrusted until a cold review finds nothing in it), and neither generated the original insight. The capability that carried the project was the *loop* — author, cold-review, repair, cross-check, repeat, with a human steering — not either model alone. We flag this because “the reviewing model is better” is exactly the self-serving inference a model writing its own project’s overview should distrust.

Where Claude held its own. Implementation work—translating an architectural idea into running Python, debugging against an independent reasoner, building cross-validation harnesses, keeping the tests green across thousands of generated concepts—is closer to ordinary programming, at which current models are strong. Documentation and audit-trail discipline (per-round superseded-paper markers, the OUTDATED archive, self-contained verification scripts) favoured patient editorial labour over insight. *Cold review across context, and across model*, was effective: a fresh frontier-model session, given a finished manuscript with no prior project context, finds defects that no session holding the long context can find. This was confirmed six times on this project—in April 2026 (Claude Opus 4.7 reviewed Claude’s own cover-tree implementation and identified the twelve-counterexample family that forced the (CT5) role-path compatibility addition, Remark 6.6); on May 28 evening (Claude Opus 4.7 reviewed GPT-5.5’s round-7 sketch and identified the three architectural defects that motivated round-8, §7); and on May 28 late evening (Claude Opus 4.8 reviewed GPT-5.5’s round-8 proof and found the D-1 completeness gap, §7.8 — and in this instance also supplied a candidate repair); and on May 29 (a fresh GPT-5.5 session reviewed GPT-5.5’s own round-9 proof and found a dual descendant-tower defect plus two foundational gaps, §7.10, which led to round-10 and the switch to the automata route); and on May 30 (a fresh GPT-5.5

session reviewed the automata proof itself and found it had the same finite-representation keystone gap, which GPT-5.5 then repaired with product-state representatives and an A/B/C companion); and on May 31 (a fresh Opus 4.8 session, a different model lineage from the proof’s author, reviewed that repair’s finite-abstraction keystone, Theorem B, and found its interface-transformer coherence asserted rather than proved, which GPT-5.5 repaired in round-12 by replacing the transformer layer with a patchwork-bag abstraction and a two-way automaton). The reviewer-was-cold property is the load-bearing one, not the reviewer-was-different-model property. And steering the cross-validation pipeline—knowing which stress tests would discriminate between candidate proofs—requires understanding both the proof and how to falsify it.

What both models did well. Long-context coherence across 30+ page proofs and many editing rounds was qualitatively different from the previous LLM generation. Tool use (Python, $\text{\LaTeX}$, git, file management, search) was routine. Seven-round adversarial dialogue between Claude and GPT-5.5—each model reviewing the other’s drafts, producing counterexamples, naming defects, proposing fixes—worked as substantive technical exchange. Hallucination was not the dominant failure mode in this project; over-elaboration was.

What both models did poorly. Original mathematical insight was the human contribution. Without Wessel’s whiteboard sketch (Figure 1) the project would not have had a target architecture to formalise. Both models also made architectural errors in their own drafts that the other caught; neither caught them in self-review. And neither reliably calibrated the cost of a fix: both round-4 (M6) verticalization and round-5/6 (V11)–(V15) were locally reasonable responses that turned out, on review, to be over-elaborated.

What this means in practice. Frontier LLMs in May 2026 can take a non-trivial open problem in mathematical logic and, over weeks of iteration with a human director and adversarial cross-review between models, produce a candidate proof together with executable artefacts that the community can verify, refute, or improve. They do not replace human expert review, and they do not generate the original intuition. But the round-9 proof is qualitatively different from what any one of the three participants would have produced alone, and the cross-validation campaign that surrounds it is qualitatively different from what would have been feasible without the implementation speed and tireless test-writing the LLMs brought to the project.

A methodology that worked: cold review as a standard intermediate step. The cycle that drove this project forward was repeatable. A fresh session of a frontier model, given the latest manuscript with no prior project context, finds defects that no session holding the long context can find. The pattern was confirmed six times (April 2026, the cover-tree implementation review that produced the (CT5) addition; May 28 evening, the round-7 review that produced round 8; May 28 late evening, the Opus 4.8 review that found the round-8 D-1 completeness gap and proposed a repair, which GPT-5.5 then ratified and integrated as round-9; and May 29, a fresh GPT-5.5 review of round-9 itself, which found a dual defect and two foundational gaps and produced round-10 and the switch to the automata route; and May 30, a fresh GPT-5.5 review of the automata proof itself, which found the same finite-representation keystone gap and was repaired with product-state representatives and an A/B/C companion; and May 31, a fresh genuinely-cold Opus 4.8—a different model lineage from the proof’s author—review of that repair’s finite-abstraction keystone (Theorem B), which found its interface-transformer coherence asserted rather than proved and was repaired in round-12 by replacing the transformer layer with a patchwork-bag abstraction and a two-way automaton). The cost is one session; the pay-off is a list of concrete defects — and, in the third instance, a candidate repair — that the long-context author can then ratify or replace. We recommend cold review as a standard intermediate step between “*this manuscript looks finished*” and “*submit for human review*”—not as a substitute for human peer review, but as a check that reliably finds real defects when there are real defects to find. The sobering corollary, visible across

all six instances, is that *each* cold review found a genuine defect in a manuscript its authors believed was finished, including successive rounds each advertised as “canonical”: the moment to trust a “finished” claim is after a cold review finds *nothing*, which has not yet happened—round-12 (the patchwork-bag proof) has not itself been cold-reviewed. The pattern is operationalisable for any project of this scale; it is not specific to description logic or to these particular models.

This is a discussion piece, not a settled result. The work has not been peer-reviewed by human experts; the primary statement (since 29 May 2026) is the split-forest + parity-tree-automata proof (§6.4), reached after the no-automata thread’s round-9 was cold-reviewed and its round-10 repairs reduced to automaton-shaped lemmas; that automata proof was itself then cold-reviewed (the 5th cold review), found to share the keystone gap, and repaired; its finite-abstraction keystone (Theorem B) was then cold-reviewed in turn (the 6th cold review) and found unproved, prompting GPT-5.5’s round-12 patchwork-bag repair (global RCC5 consistency via the patchwork property, decided by a two-way parity automaton), which has not itself been cold-reviewed or machine-certified. The decidability theorem is strongly supported but not certified. The meta-research point—*that this kind of work is now possible, and that the result is worth engaging with on its technical merits*—is the more important contribution. We invite the description logic community to engage with both layers.

8.6. A note on citation verification

Because AI systems are known to hallucinate bibliographic references, all external citations in this paper and its companion papers were verified via web search against journal landing pages, CEUR-WS/LIPICs/arXiv metadata, and indexed proceedings. The audit caught one fully hallucinated reference (the non-existent “Rybina–Zakharyashev 2001”—the genuine paper at that venue is Reynolds and Zakharyashev, *On the products of linear modal logics*, JLC 11(6), 2001) and several transcription errors (incorrect coauthors, page ranges, venues, and journal names). All fixes are documented in the project’s CONVERSATION.md log with commit hashes. Citations in this paper have been checked, but readers should still verify any reference before citing it.

Declaration on Generative AI

In preparing this work, the author (Wessel) made significant use of generative-AI language tools: **Claude** (Anthropic, Opus 4.7 and Opus 4.8), **GPT-5.4 Pro** (OpenAI), and **GPT-5.5** (OpenAI). Specifically, the AI tools were used to:

- formalise the mathematical proofs (split-forest semantics, completeness extraction across nine review rounds, the round-7 pair-shape / incidence-tag architecture, the round-8 saturated-side-context construction, and the round-9 composition-forced verticalization that closes the D-1 gap);
- perform the cold reviews that found successive defects (including the round-8 completeness gap D-1) and draft the proposed repairs, which the proof author then ratified and integrated (round-9);
- implement the cover-tree tableau decision procedure (≈ 350 lines of Python), the quasimodel reasoners, and the WP1–WP9 verification scripts;
- conduct computational verification campaigns (911-concept cross-validation, 775-concept decomposition test, the round-7 stress-concept verifications, the round-8/round-9 composition-arithmetic checks);
- draft the $\text{\LaTeX}$ manuscript text, including this overview and all companion papers in the repository.

The division of labour between the human author and the AI tools is documented in detail in §8.5 of this paper.

Responsibility. By signing this work as its author, Wessel takes full responsibility for all of its contents, irrespective of how the contents were generated. Where the generative-AI tools produced inappropriate language, plagiarised content, biased content, errors, mistakes, incorrect references, or misleading content that was included in the work, that responsibility rests with the author, not with the tools.

Authorship. Per arXiv’s content moderation policy, generative-AI tools are not listed as authors of this paper. The AI tools are disclosed here as the substantial methodological instruments they were.

Direction. The research direction—problem formulation, the key conceptual insights (cover-tree intuition, PPI-tree orientation, EQ-splitting, DR-rigidity), the correction of architectural errors, and the high-level judgement calls about when to converge—was provided by Wessel; the AI tools performed the detailed formalisation, implementation, verification, cold review, and manuscript drafting under that direction.

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